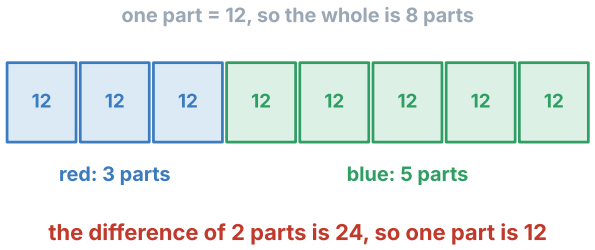
# SAT Math 3.1 — Ratios, Rates, Proportion & Units

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Ratios — part to part, or part to whole?

**a : b means a parts and b parts, so the whole is a + b parts**

'Boys to girls is 3 : 5' means boys are **3/8** of the class, not 3/5. **Add the parts first**, then take the share you need. When a question gives a **difference** instead of a total, count how many parts that difference is worth.



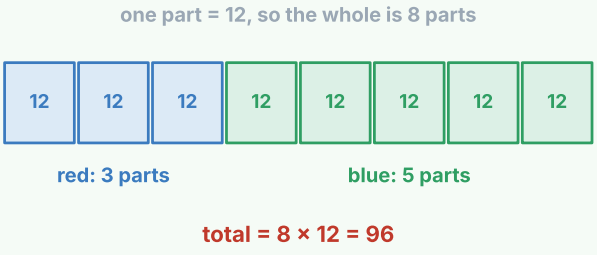

Eight equal parts in all: three red and five blue. The gap between them is two parts, which is what a 'how many more' question gives you.

**Example 1.** **Red and blue counters are in the ratio 3 : 5 and there are 24 more blue than red. How many counters altogether?**

**Solution.**

The difference is 5 − 3 = 2 parts, so 2 parts = 24 and one part = 12.

Total = 8 parts = **96**.

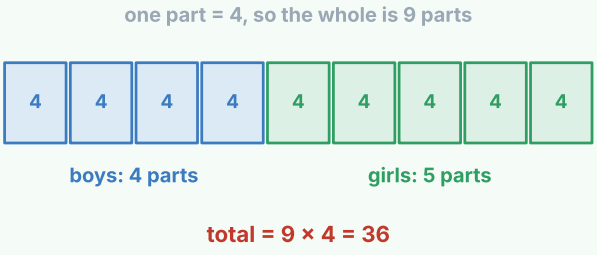



**Example 2.** **A class of 36 has boys and girls in the ratio 4 : 5. How many girls?**

**Solution.**

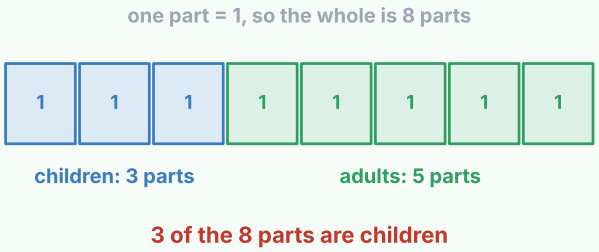
9 parts, so one part is 4.

Girls = 5 × 4 = **20**.



**Example 3.** **Three-eighths of a group are children. What is the ratio of children to adults?**

**Solution.**
Children 3 parts of 8, so adults are 8 − 3 = 5.
The ratio is **3 : 5**.




## 2 Proportion, and direct against inverse

**direct: y = kx — double one, double the other | inverse: y = k/x — double one, halve the other**

Decide which you have before calculating: more items cost more (**direct**), more workers take fewer days (**inverse**). Inverse questions are easiest as a fixed total of work — 5 painters × 12 days = 60 painter-days.

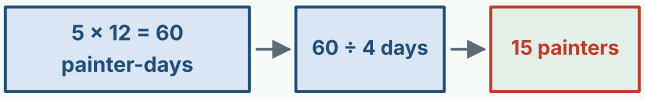
name the type before you calculate

| | |
|---|---|
| direct: y = kx | twice the items, twice the cost |
| inverse: y = k/x | twice the workers, half the days |
| the test does more of one mean more, or less, of the other? | |

The two behave in opposite directions, and the words in the question tell you which is which.

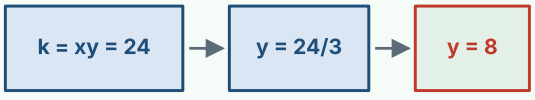
**Example 1.** **5 painters paint a house in 12 days. How many painters are needed to finish it in 4 days?**

**Solution.**
Total work = 5 × 12 = 60 painter-days.
60 ÷ 4 = **15 painters**. Fewer days needs more painters — inverse.



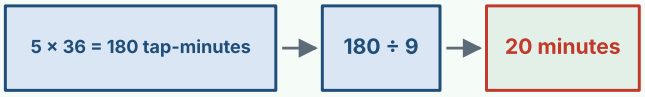

**Example 2.** **y is inversely proportional to x, and y = 6 when x = 4. Find y when x = 3.**

**Solution.**
k = xy = 24, so y = 24/3 = **8**. x went down, y went up.



**Example 3.** **The time to fill a tank is inversely proportional to the number of taps. 5 taps take 36 minutes. How long do 9 taps take?**

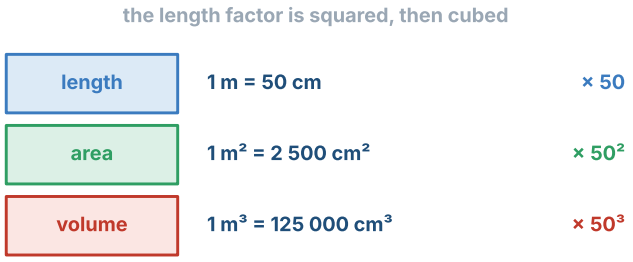
**Solution.**
Total work = 5 × 36 = 180 tap-minutes.
180 ÷ 9 = **20 minutes**.




## 3 Scale drawings and maps

**a scale of 1 : n means 1 unit on the drawing is n of the same units in reality**

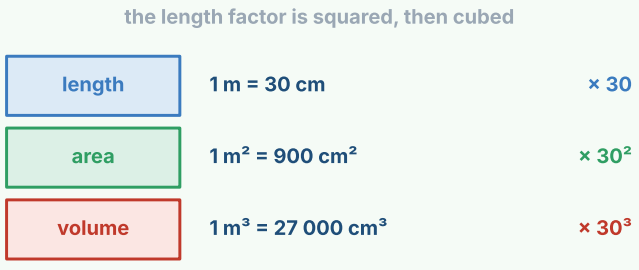
1 : 50,000 means one centimetre is 50,000 **centimetres**. Multiply first, **then** convert. And if lengths scale by k, **areas scale by $k^2$** and **volumes by $k^3$**.




The same factor applies once to a length, twice to an area and three times to a volume — which is why a 1 : 50 model's floor area is 2,500 times smaller, not 50.

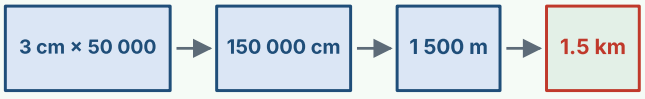
**Example 1.** **A model is built to scale 1 : 30 and holds 2 litres. How much does the real object hold?**

**Solution.**
Volumes scale by $30^3$ = 27,000.
2 × 27,000 = **54,000 litres** — not 60.




**Example 2.** **A map has scale 1 : 50,000. A distance of 3 cm on the map is how many km?**

**Solution.**
3 × 50,000 = 150,000 cm = 1,500 m.
**1.5 km.** Multiply first, convert second.



**Example 3.** **On a 1 : 50 model the floor area is 1,200 cm². Find the real area in m².**

**Solution.**
Areas scale by $50^2 = 2{,}500$.
$1{,}200 \times 2{,}500 = 3{,}000{,}000$ cm² = **300 m²**.

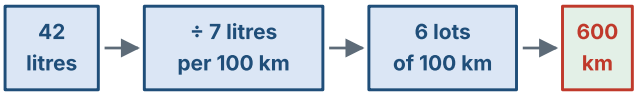
the length factor is squared, then cubed

| | | |
|---|---|---|
| length | 1 m = 50 cm | × 50 |
| area | 1 m² = 2 500 cm² | × 50² |
| volume | 1 m³ = 125 000 cm³ | × 50³ |

## 4 Unit rates and unit conversion

**a unit rate is the amount per one | convert by multiplying by a fraction equal to 1, arranged so the unwanted unit cancels**

The word after 'per' goes on the bottom. Write a conversion as a fraction with the unit you want on top and the unit you are losing underneath, and **never stop at the number** — check you finished in the unit asked for.




the question asked for a distance, not a count of lots

Six is the trap: it is the number of 100-km lots, not the distance. One more step finishes the job.

**Example 1.** **A car uses 7 litres of fuel per 100 km. How far can it travel on 42 litres?**

**Solution.**
$42 \div 7 = 6$ lots of 100 km → **600 km**.
Stopping at 6 is the trap: the question asked for a distance, not a count.




**Example 2.** **Which is better value: 400 g for $3.20, or 650 g for $4.55?**

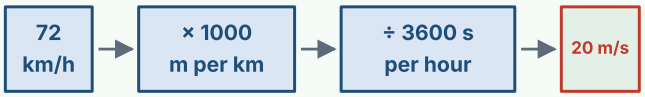
**Solution.**
$8.00 per kg against **$7.00 per kg**, so the 650 g pack. Convert both to the same rate first.

| | |
|---|---|
| 400 g for $3.20 | $8.00 per kg |
| 650 g for $4.55 | $7.00 per kg — better |

**Example 3.** **Convert 72 km/h into metres per second.**

**Solution.**

72 × 1,000 ÷ 3,600 = **20 m/s**. Kilometres and hours both cancel.




## 5 Squared and cubed units, density and flow

**1 m² = 10,000 cm² · 1 m³ = 1,000,000 cm³ · 1 m³ = 1,000 L | density = mass ÷ volume**

Dividing an area by 100 instead of 10,000 is the error this box exists to prevent. Flow rates **add** when two taps run together; times never do.

the length factor is squared, then cubed

| | | |
|---|---|---|
| length | 1 m = 100 cm | × 100 |
| area | 1 m² = 10 000 cm² | × 100² |
| volume | 1 m³ = 1 000 000 cm³ | × 100³ |

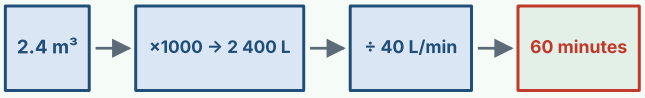
One hundred centimetres to the metre, so ten thousand square centimetres to the square metre and a million cubic ones. Each row is the one above, multiplied by 100 again.

**Example 1.** **A tank measures 2 m by 1.5 m by 0.8 m and fills at 40 litres per minute. How long does it take?**

**Solution.**

Volume = 2.4 m³ = **2,400 litres**.

2,400 ÷ 40 = **60 minutes**. Two unit steps, not one.




**Example 2.** **A material has density 2.7 g/cm³. Express this in kg/m³.**

**Solution.**

× 1,000,000 for the volume, ÷ 1,000 for the mass.

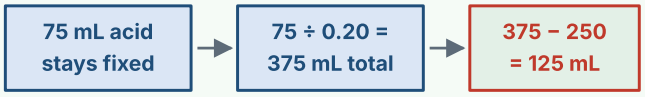
**2,700 kg/m³**.

the length factor is squared, then cubed

| | | |
|---|---|---|
| length | 1 m = 100 cm | × 100 |
| area | 1 m² = 10 000 cm² | × 100² |
| volume | 1 m³ = 1 000 000 cm³ | × 100³ |

**Example 3.** **250 mL of a solution is 30% acid. How much water makes it 20% acid?**

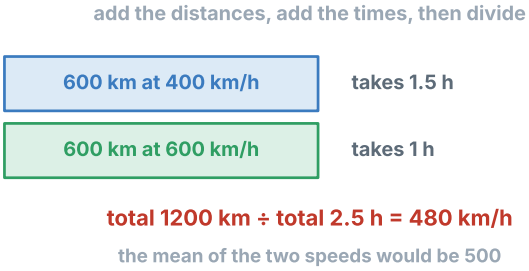
**Solution.**
Acid is 75 mL and does not change. 75 ÷ 0.20 = 375 mL in total.
Water added = **125 mL**.




## 6 Average speed — total over total

**average speed = total distance ÷ total time (never the average of the speeds)**

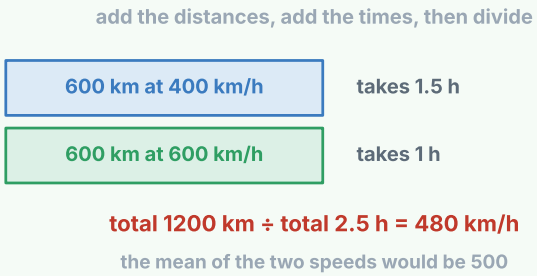
This was the worst-answered question on your class's mock: 2 of 15. Averaging the two speeds is only correct when the legs take the **same time**, which is almost never what the question describes.




The slow leg takes longer, so it carries more weight. That is why the answer sits below the halfway point between 400 and 600.

**Example 1.** **A plane flies 600 km at 400 km/h and returns at 600 km/h. Find the average speed.**

**Solution.**
Times 1.5 h and 1 h, so 2.5 h in total for 1,200 km.
1,200 ÷ 2.5 = **480 km/h** — not 500.

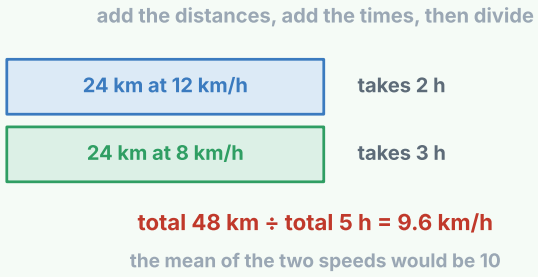


**Example 2.** **A cyclist rides 24 km at 12 km/h, then 24 km at 8 km/h. Find the average speed.**

**Solution.**

Times 2 h and 3 h, so 5 h for 48 km.

48 ÷ 5 = **9.6 km/h** — not 10.

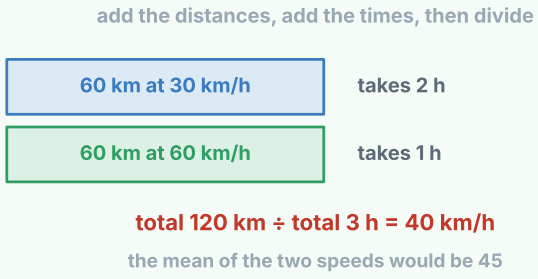



**Example 3.** **Half a journey is covered at 30 km/h and half at 60 km/h. Find the average speed.**

**Solution.**

For a distance 2d the time is d/30 + d/60 = d/20.

2d ÷ (d/20) = **40 km/h**, not 45.



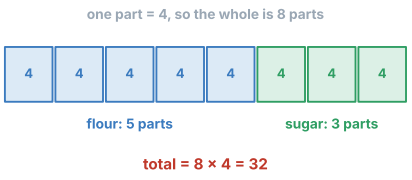
## Part 1 — With Desmos (10 questions · every one is faster on the graph)

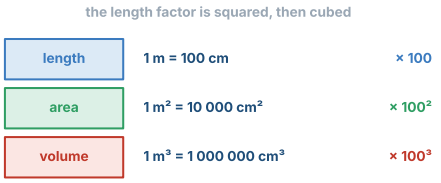
The SAT gives you Desmos on every maths question. For this topic it is **an equation solver and a unit chain**. Put the whole conversion on one line so no step is lost, and type a relationship with an x in it when the question hides one.

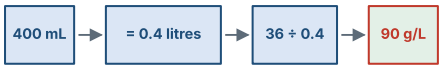
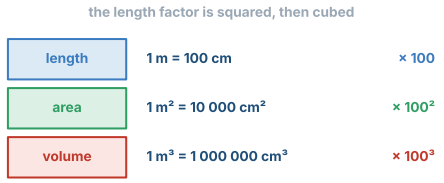
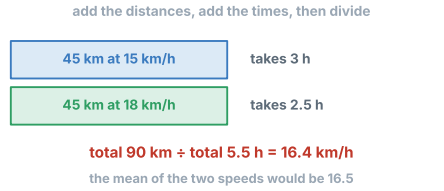
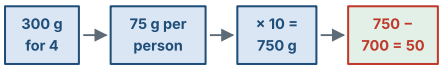
| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A recipe uses flour and sugar in the ratio 5 : 3. If 20 cups of flour are used, how many cups of sugar are needed? | `20*3/5` | Scale the ratio in one line. | 12 |
| D2 | A car travels 240 kilometres in 3 hours at a constant speed. What is its speed in kilometres per hour? | `240/3` | Distance over time. | 80 |
| D3 | A printer produces 18 pages in 40 seconds. At this rate, how many pages does it produce in 5 minutes? | `18/40*300` | Convert minutes to seconds inside the line. | 135 |
| D4 | A map uses a scale of 1 cm to 25 km. Two towns are 7.4 cm apart on the map. How many kilometres apart are they? | `7.4*25` | Multiply by the scale factor. | 185 |
| D5 | A rectangular tank measures 40 cm by 25 cm by 30 cm. How many litres does it hold? (1 litre = 1000 $cm^3$) | `40*25*30/1000` | Divide by 1000, not by 1,000,000 — that factor converts to cubic metres. | 30 |
| D6 | A solution contains 36 grams of salt in 400 millilitres. What is the concentration in grams per litre? | `36/0.4` | Convert millilitres to litres first. | 90 |
| D7 | A machine fills 450 bottles in 1 hour 15 minutes. How many bottles per minute is that? | `450/75` | Work in a single unit before dividing. | 6 |
| D8 | A metal block has a mass of 540 grams and a volume of 60 cubic centimetres. What is its density, in grams per cubic centimetre? | `540/60` | Density is mass over volume. | 9 |
| D9 | A shop sells 3 notebooks for $7.50. At the same rate, what do 11 notebooks cost, in dollars? | `7.50/3*11` | Find the unit price, then scale. | 27.5 |
| D10 | A runner covers 12 kilometres in 50 minutes. What is this speed in kilometres per hour? | `12/50*60` | Scale the time unit up to an hour. | 14.4 |

## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two routes in: if the question hides a **relationship**, write an equation; if it hides a **unit**, write the conversion as a fraction and let the units cancel.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | A recipe calls for flour and sugar in the ratio 5 : 3. A baker uses 20 cups of flour. How many cups of sugar should the baker use?   | 12 | Each part is 20 ÷ 5 = 4 cups, and sugar is 3 parts.<br>3 × 4 = **12 cups**. A common wrong answer is the difference between the parts, not the sugar. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P2 | Hard | A printer produces 18 pages in 40 seconds. At this rate, how many pages will it produce in 5 minutes?<br>  | 135 | 5 minutes is 300 seconds, and the rate is 18/40 = 0.45 pages per second. 0.45 × 300 = **135 pages**. A common wrong answer comes from using 5 minutes as 200 seconds. |
| P3 | Hard | A rectangular oil tin measures 40 cm by 25 cm by 30 cm and is completely full. How many litres of oil does it hold? (1 litre = 1000 cubic centimetres)<br>  | 30 | Volume = 40 × 25 × 30 = 30,000 cm³. 30,000 ÷ 1000 = **30 litres**. Dividing by 1,000,000 gives 0.03 — that factor converts cm³ to *cubic metres*, not litres. |
| P4 | Hard | On a map, 1 centimetre represents 25 kilometres. Two towns are 7.4 centimetres apart. What is the actual distance between them, in kilometres?<br>  | 185 | 7.4 × 25 = **185 km**.<br>A common wrong answer divides instead of multiplying; another adds the two numbers. |
| P5 | Hard | A metal block has a mass of 540 grams and a volume of 60 cubic centimetres. What is its density, in grams per cubic centimetre?<br>  | 9 | Density = mass ÷ volume = 540/60. = **9 g/cm³**. A common wrong answer inverts the fraction, giving volume per gram instead. |
| P6 | Hard | A machine fills 450 bottles in 1 hour and 15 minutes. What is its rate, in bottles per minute?<br>  | 6 | 1 hour 15 minutes is 75 minutes. 450 ÷ 75 = **6 bottles per minute**. A common wrong answer divides by 60, forgetting the extra 15 minutes. |
| P7 | Very hard | A runner covers 12 kilometres in 50 minutes. What is this speed in kilometres per hour?<br>  | 14.4 | In one minute the runner covers 12/50 = 0.24 km, and an hour is 60 minutes. 0.24 × 60 = **14.4 km/h**. A common wrong answer divides 12 by 0.5, treating 50 minutes as half an hour. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P8 | Very hard | A solution contains 36 grams of salt dissolved in 400 millilitres of water. What is the concentration, in grams per litre?<br>  | 90 | 400 mL = 0.4 L.<br>36 ÷ 0.4 = **90 g/L**. A common wrong answer divides by 4 instead of 0.4 — a factor-of-ten slip the units would have caught. |
| P9 | Very hard | Three identical notebooks cost $7.50. A school buys 44 notebooks and receives a 10% discount on the total. How much does the school pay, in dollars?<br>  | 99 | Unit price = 7.50/3 = $2.50, so 44 cost 44 × 2.50 = $110.<br>A 10% discount means paying 90%: 0.90 × 110 = **$99**. |
| P10 | Very hard | A cube has edges of 1 metre. A student says its volume is 100 cubic centimetres because 1 m = 100 cm. What is the correct volume, and what went wrong?<br>  | 1,000,000 cm³; the conversion factor must be cubed. | A length factor of 100 becomes 100² for area and 100³ for volume.<br>1 m³ = 100³ = **1,000,000 cm³**. This is the same slip that turns 30 litres into 0.03. |
| P11 | Very hard | Two cyclists ride at constant speeds. The first covers 45 km in 3 hours; the second covers the same distance in 2 hours 30 minutes. How much faster is the second, in kilometres per hour?<br>  | 3 | First: 45/3 = 15 km/h. Second: 2 h 30 min = 2.5 h, so 45/2.5 = 18 km/h.<br>18 − 15 = **3 km/h**. A common wrong answer gives the second speed itself, not the difference. |
| P12 | Very hard | A recipe for 4 people uses 300 grams of rice. A cook prepares the same recipe for 10 people but has only 700 grams of rice. How many grams short is the cook?<br>  | 50 | For 10 people: 300 ÷ 4 = 75 g per person, and 75 × 10 = 750 g.<br>750 − 700 = **50 grams short**. |

**The one habit to build.** Eight of these twelve finish one step *before* the answer: 2.4 m³ is 2,400 litres, 6 lots of 100 km is 600 km, 48/5 is 9.6 and not 10. Every time you reach a number ask: *is this what they asked for, in the unit they asked for?*

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*

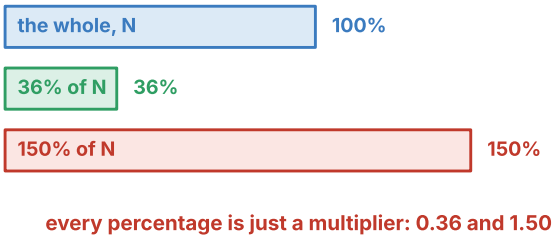
# SAT Math 3.2 — Percentages

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Percent of a number

**p% of N = (p/100) × N — the word *of* always means multiply**

Turn the percentage into a decimal and multiply. Percentages above 100 are **not special**: 150% of 60 is 1.5 × 60 = 90. Run it backwards with one division — if 12% of x is 54, then x = 54 ÷ 0.12.




A percentage above 100 sits past the whole bar — nothing about the method changes, only the multiplier.

**Example 1. A shop has 1,250 items. 36% are books, and 20% of the books are hardback. How many hardback books are there?**

**Solution.**

0.36 × 1,250 = **450 books**.

0.20 × 450 = **90 hardback books**.




**Example 2. A shop has 1,250 items. 36% are books, and 20% of the books are hardback. How many hardbacks?**

**Solution.**

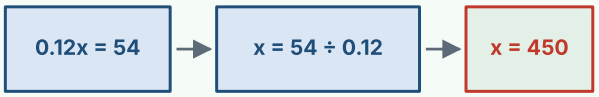
0.36 × 1,250 = 450 books; 0.20 × 450 = **90**.

the second percentage is taken of the books, not of 1,250

| | |
|---|---|
| 0.36 × 1,250 | 450 books |
| 0.20 × 450 | 90 hardbacks |

**Example 3.** **If 12% of x is 54, what is x?**

**Solution.**
0.12x = 54 → x = 54 ÷ 0.12 = **450**.




## 2 Increase and decrease — multiply by what is left

**up p%: × (1 + p/100) | down p%: × (1 − p/100)**

One multiplication, **not a subtraction**. To take 25% off, multiply by 0.75; to add 15%, multiply by 1.15. Writing the multiplier straight down is what makes every harder question in this topic a one-liner.

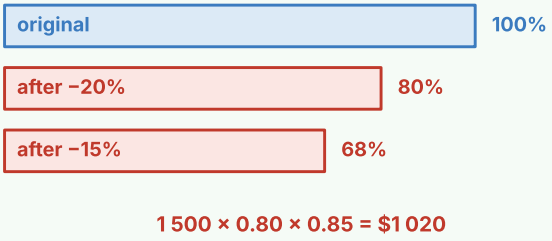
the percentage is the distance from 1, never the base

| | |
|---|---|
| rise of 15% → | b = 1 + 0.15 = 1.15 |
| rise of 40% → | b = 1 + 0.40 = 1.40 |
| fall of 20% → | b = 1 − 0.20 = 0.80 |
| fall of 25% → | b = 1 − 0.25 = 0.75 |

The percentage is the distance from 1. What you multiply by is what is left, or what you end up with — never the percentage itself.

**Example 1.** **A $1,500 laptop falls 20% in its first year and a further 15% in its second. Find the final price.**

**Solution.**
1,500 × 0.80 × 0.85 = **$1,020**.
Multiplying by 0.25 to take 25% off is the classic slip — the multiplier is what is *left*.



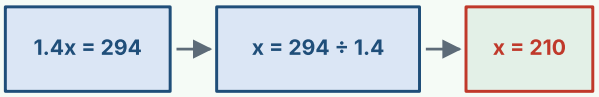

**Example 2.** **A $1,500 laptop falls 20% in year one and a further 15% in year two. Find the final price.**

**Solution.**
1,500 × 0.80 × 0.85 = **$1,020**.



**Example 3.** **A number increased by 40% gives 294. Find the number.**

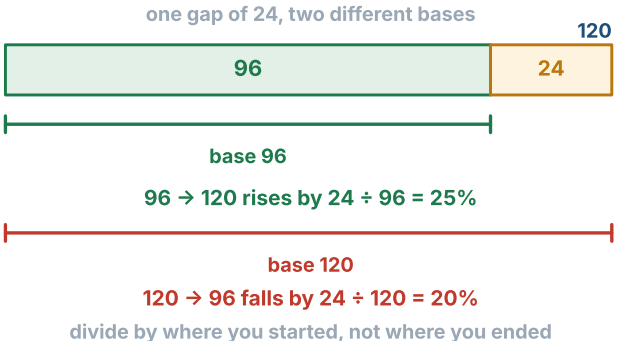
**Solution.**
1.4x = 294 → x = **210**. Divide by the multiplier.




## 3 Percent change, and one quantity as a percent of another

**percent change = (new − old) ÷ old × 100 | A as a percent of B = (A ÷ B) × 100**

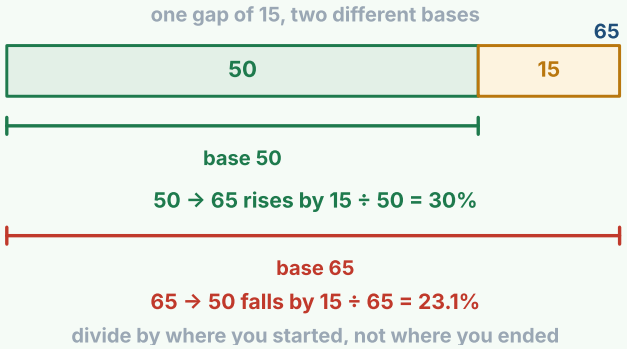
The denominator is always **what you started from**, or whatever follows the word *of*. The same gap is a different percentage in each direction: 96 → 120 is a 25% rise, but 120 → 96 is a 20% fall.




One gap of 24, two answers. Measured against 96 it is a quarter; measured against 120 it is a fifth. The base decides.

**Example 1.** **A ticket price rises from $50 to $65. What is the percent increase?**

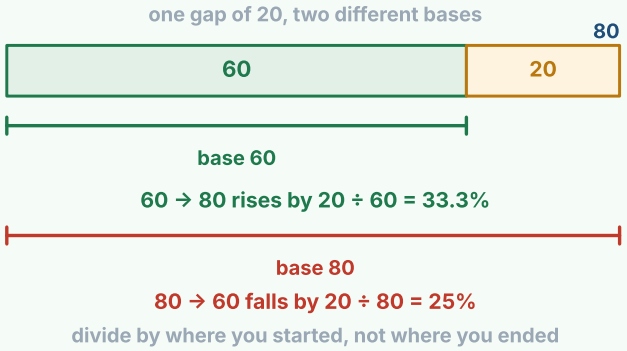
**Solution.**
(65 − 50)/50 = 15/50 = **30%**.
Dividing by 65 gives about 23% — the right subtraction and the wrong denominator.



**Example 2.** **A value falls from 80 to 60. Find the percent decrease.**

**Solution.**

20 ÷ 80 = **25%**. The denominator is the old value.

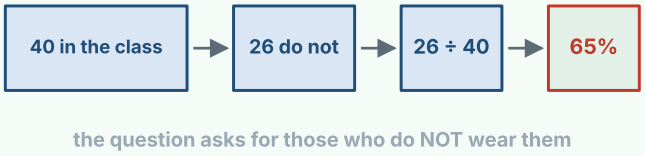



**Example 3.** **In a class of 40, 14 students wear glasses. What percent do not?**

**Solution.**

40 − 14 = 26 students.

26 ÷ 40 = **65%**.



## 4 Reverse percent — finding the original

**if the final value is F after a change of p%, then original = F ÷ (1 ± p/100)**

Given the price **after** a change, you divide. Adding the percentage back on is the trap: 15% off $120 is $102, but 15% more than $102 is only $117.30. Write 0.85x = 102 and solve it.




**102 ÷ 0.85 = 120**

**one division by the whole chain — never add the percentage back on**

The arrow runs forwards, so reading it backwards means dividing. Adding 15% of $102 back on lands at $117.30 — the wrong base.

**Example 1. An item sells for $135 including 8% tax. What was the pre-tax price?**

**Solution.**

1.08x = 135, so x = 135 ÷ 1.08 = **$125**.

Tax is *added*, so divide by 1.08 — not by 0.92.




**135 ÷ 1.08 = 125**

**one division by the whole chain — never add the percentage back on**

**Example 2. A price is $288 after 20% off and then a further 10% off. Find the original.**

**Solution.**

0.8 × 0.9 = 0.72, so 288 ÷ 0.72 = **$400**. Divide by the whole chain.



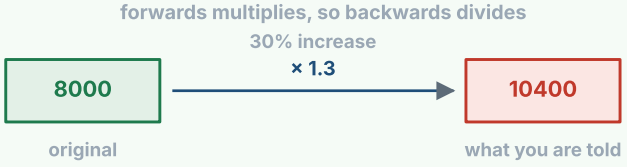

**288 ÷ 0.72 = 400**

**one division by the whole chain — never add the percentage back on**

**Example 3. After a 30% increase a population is 10,400. What was it before?**

**Solution.**

10,400 ÷ 1.3 = **8,000**.



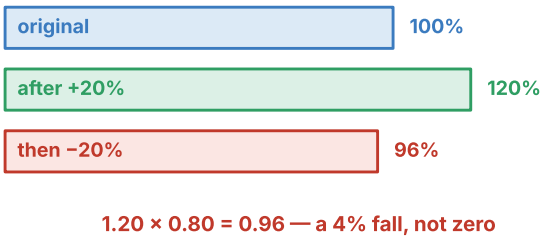

**10400 ÷ 1.3 = 8000**

**one division by the whole chain — never add the percentage back on**

## 5 Successive percent change — multiply, never add

**up p% then down q% → multiply by (1 + p/100)(1 − q/100)**

Two changes **do not add**, because the second is taken from a bigger or smaller number. The recovery is always larger than the fall: down 20% needs a 25% rise to get back, since 0.8 × 1.25 = 1.




The second 20% is taken from 120, not from 100, so it removes more than the first 20% added. The bar lands below where it started.

**Example 1. A price rises 20% and then falls 20%. What is the overall change?**

**Solution.**
1.20 × 0.80 = 0.96, so the price is 96% of the original — a **4% decrease**, not zero.
The order makes no difference: 0.80 × 1.20 is the same 0.96.



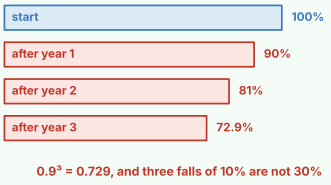

**Example 2. A price falls 20%. By what percent must it rise to restore the original?**

**Solution.**
0.8 × m = 1 → m = 1.25, a **25% rise**. The recovery is always bigger than the fall.




**Example 3. A value falls 10% each year for three years. It ends at what percent of the original?**

**Solution.**
$0.9^3 = 0.729$, so **72.9%**. Repeated change is a power.



## 6 Discount, markup, tax and commission

**markup is on the cost · discount is on the marked price · tax is on the price after the discount · percent profit is measured on the cost**

The order is **not decoration** — doing tax before the discount gives a different, wrong answer. Commission is often tiered or applies only above a threshold, so read **which sales it covers**.

each percentage has its own base — work left to right

| | |
|---|---|
| markup | taken on the cost |
| discount | taken on the marked price |
| tax | taken on the price after the discount |
| percent profit | measured against the cost |

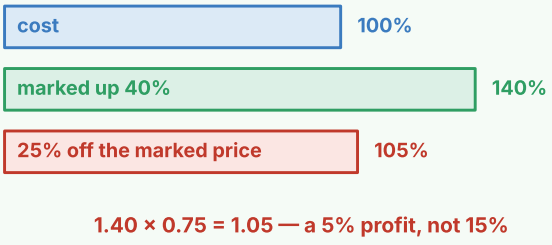
Four percentages, four different bases. Reading the chain in order is the whole method.

**Example 1. A shop marks goods up 40% above cost and then offers 25% off. What percent profit is made on the cost?**

**Solution.**

1.40 × 0.75 = 1.05, so the selling price is 105% of cost — a **5% profit**.

Not 15%. The 25% came off the *marked price*, not the cost.




**Example 2. A $250 coat is reduced 30%, then 8% tax is added to the sale price. Find the total.**

**Solution.**

250 × 0.7 = $175, then 175 × 1.08 = **$189**. Tax is on what you actually pay.

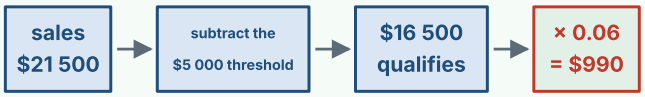



tax after the discount, never before

**Example 3.** **Commission is 6% on sales above $5,000 only. Find the commission on sales of $21,500.**

**Solution.**

Only the excess counts: 0.06 × 16,500 = **$990**.



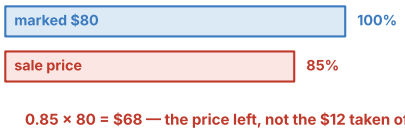
## Part 1 — With Desmos (10 questions · every one is faster on the graph)

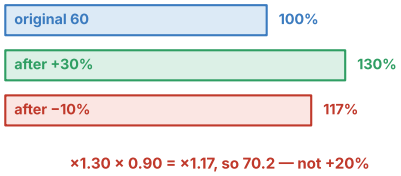
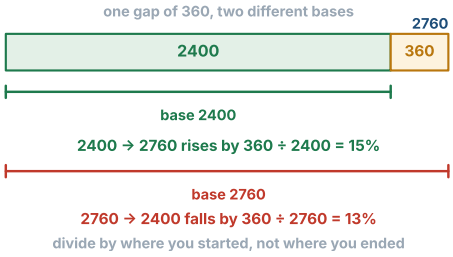
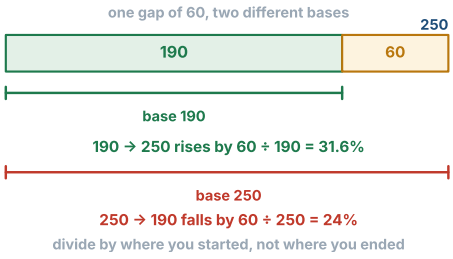
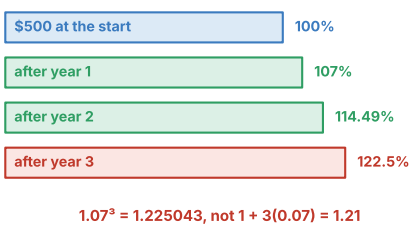
The SAT gives you Desmos on every maths question. For percentages it is **a multiplier chain**. Type every multiplier onto one line and read the final number; where the original is unknown, type the equation and let Desmos solve it.

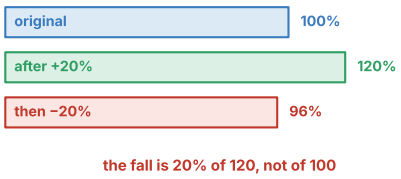
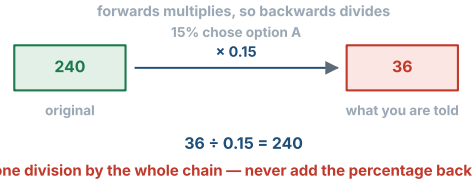
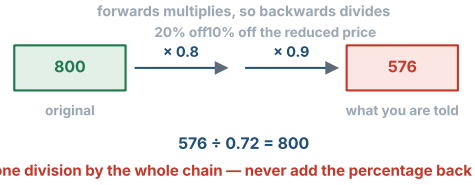
| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A jacket priced at $80 is reduced by 15%. What is the sale price, in dollars? | `80*0.85` | A 15% cut multiplies by 0.85. | 68 |
| D2 | A salary of $42,000 rises by 6%. What is the new salary, in dollars? | `42000*1.06` | A 6% rise multiplies by 1.06. | 44520 |
| D3 | After a 15% discount an item sells for $61.20. What was the original price, in dollars? | `61.20/0.85` | Reversing a discount means dividing. | 72 |
| D4 | A price of 60 is raised by 30% and the new price is then cut by 10%. What is the final price? | `60*1.3*0.9` | Chain the multipliers; the answer is not 60 × 1.20. | 70.2 |
| D5 | A population grows from 2,400 to 2,760. What is the percentage increase? | `(2760-2400)/2400*100` | Change over the ORIGINAL, times 100. | 15 |
| D6 | What is 18% of 350? | `0.18*350` | Write the percentage as a decimal. | 63 |
| D7 | A $90 item is taxed at 8%. What is the total cost, in dollars? | `90*1.08` | Adding tax multiplies by 1 + rate. | 97.2 |
| D8 | A number decreases from 250 to 190. What is the percentage decrease? | `(250-190)/250*100` | Still divide by the original value. | 24 |
| D9 | A $500 investment grows by 7% a year for 3 years. What is it worth, in dollars? | `500*1.07^3` | Repeated multiplication, not 500 × 1.21. | 612.52 |
| D10 | 36 is what percentage of 240? | `36/240*100` | Part over whole, times 100. | 15 |

## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two routes in: if a percentage is running **backwards**, divide; if two changes are **stacked**, multiply. And check which number the percentage is taken *of*.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | A jacket is priced at $80. During a sale the price is reduced by 15%. What is the sale price of the jacket?<br>  | $68.00 | A 15% reduction leaves 85% of the price. 0.85 × 80 = **$68**. A common wrong answer is the discount itself, not the new price. |
| P2 | Hard | After a discount of 15%, a brass scale sells for 61.20 talens. What was its original price, in talens?<br>  | 72.00 | The sale price is 85% of the original, so divide. 61.20 ÷ 0.85 = **72**. A common wrong answer adds 15% of 61.20 back on, which reverses the wrong base — the commonest error here. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P3 | Hard | A copper kettle is priced at 60 talens. The shop raises the price by 30%, then lowers the new price by 10%. What is the final price, in talens?<br>  | 70.2 | 60 × 1.30 = 78, then 78 × 0.90 = **70.2**.<br>Successive changes multiply: 1.30 × 0.90 = 1.17, a rise of 17% — not the 20% you get by adding 30 and −10. |
| P4 | Hard | A town's population grows from 2,400 to 2,760. What is the percentage increase?<br>  | 15% | Change = 360, and the base is the original 2,400.<br>360/2400 = 0.15 = **15%**. A common wrong answer divides by the new value 2,760 instead. |
| P5 | Hard | A $90 item is subject to 8% sales tax. What is the total cost, to the nearest cent?<br>  | 97.20 | 90 × 1.08 = **$97.20**. |
| P6 | Hard | A number decreases from 250 to 190. What is the percentage decrease?<br>  | 24% | Change = 60, over the original 250.<br>60/250 = 0.24 = **24%**. A common wrong answer divides by 190; D is the proportion remaining, not the decrease. |
| P7 | Very hard | A $500 investment grows by 7% each year. To the nearest cent, what is it worth after 3 years?<br>  | $612.52 | $500(1.07)^3 = 500 \times 1.225043 =$ **$612.52**.<br>A common wrong answer is 500 × 1.21, which multiplies the rate by 3 instead of compounding it. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P8 | Very hard | The price of a ticket rises by 20% and then falls by 20%. Compared with the original price, the final price is   | 4% lower. | 1.20 × 0.80 = 0.96, so the final price is 96% of the original — **4% lower**. The two 20%s are taken from different bases, which is why they do not cancel. |
| P9 | Very hard | A shop marks up the cost of an item by 40% to set its selling price. During a sale it then takes 25% off the selling price. What is the overall percentage change from the original cost?   | 5% increase | 1.40 × 0.75 = 1.05, so the final price is 105% of the cost — a **5% increase**. A common wrong answer adds 40 and −25, which treats both percentages as if they had the same base. |
| P10 | Very hard | In a survey, 36 of the respondents chose option A, which was 15% of all respondents. How many people responded in total?   | 240 | If 36 is 15% of the total, then total = 36 ÷ 0.15 = **240 people**. |
| P11 | Very hard | A laptop's price is reduced by 20%, and a customer with a membership card receives a further 10% off the reduced price. The customer pays $576. What was the original price, in dollars?   | $800.00 | The two reductions multiply: 0.80 × 0.90 = 0.72 of the original. 576 ÷ 0.72 = **$800**. A common wrong answer divides by 0.90 only; B treats the total discount as 20%. |
| P12 | Very hard | A shop's revenue rose 25% in year 1 and fell 20% in year 2. A manager reports that revenue is now 5% above where it started. Is the manager right?   | No; revenue is back to exactly where it started. | 1.25 × 0.80 = 1.00 exactly, so revenue is **unchanged**. Adding and subtracting the percentages (that answer) ignores that the 20% fall is taken from the larger, post-increase figure. |

**The one habit to build.** Every percentage is a percentage **of something**, and the SAT's whole game is changing which something. Profit is on the cost, percent change is on the old value, tax is on the discounted price. Before you divide, say out loud: *of what?*

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*

# SAT Math 3.3 — One-Variable Data: Centre & Spread

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Mean, median and mode

**mean = total ÷ how many | median = middle value once sorted | mode = most frequent**

**Sort the list before taking a median** — that single step is the whole trick. With an even count the median is the average of the two middle values. The mean uses every value, so it moves when any one of them moves; the median usually does not.



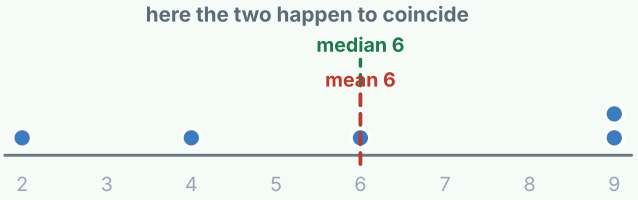

The median is a *position*, so the list has to be in order before you look at the middle. Here 6 sits third of five.

**Example 1.** **Find the mean and the median of 4, 9, 6, 2, 9.**

**Solution.**

Mean = 30 ÷ 5 = **6**.

Sorted: 2, 4, 6, 9, 9, so the median is **6**.

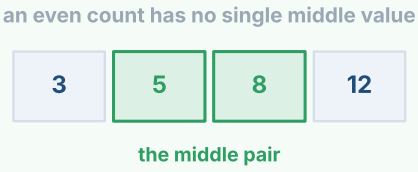



**Example 2.** **Find the median of 3, 8, 5, 12.**

**Solution.**

Sorted: 3, 5, 8, 12. Even count, so average the middle two.

(5 + 8)/2 = **6.5**.



**Example 3.** **The mean of 5 numbers is 14. What is their total?**

**Solution.**

Total = mean × how many = 14 × 5 = **70**.

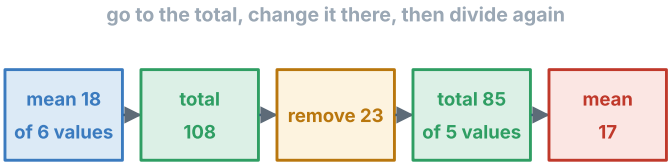
Turning a mean back into a total is the move behind most of these questions.




## 2 Changing the data set

**work with the total, not the mean — total = mean × count**

Whenever a value is added, removed or changed, **convert to the total first**, adjust it, then divide by the **new** count. Trying to patch the mean directly is where the errors come from.



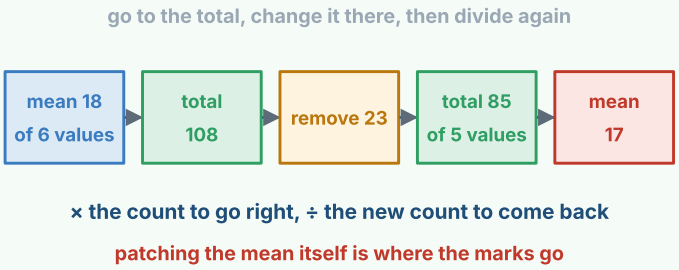

Four short steps, and the count changes in the middle of them. Dividing by the old count is the slip this picture is here to stop.

**Example 1.** **The mean of 6 numbers is 18. The number 23 is removed. Find the new mean.**

**Solution.**

Total = 108; removing 23 leaves 85 over 5 numbers.

85 ÷ 5 = **17**.

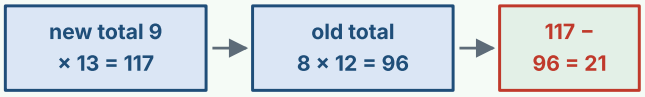


**Example 2.** **The mean of 8 numbers is 12. A ninth number is added and the mean becomes 13. Find the ninth.**

**Solution.**

New total = 9 × 13 = 117; old total = 96.

117 − 96 = **21**.




the new value is the difference between the two totals

**Example 3.** **A value of 20 in a list of 10 numbers with mean 15 is changed to 40. Find the new mean.**

**Solution.**

Total 150 rises by 20 to 170, still over 10 numbers.

170 ÷ 10 = **17**.

go to the total, change it there, then divide again




**× the count to go right, ÷ the new count to come back**

**patching the mean itself is where the marks go**

## 3 Weighted average

**weighted mean = (sum of value × weight) ÷ (sum of weights)**

When groups are different sizes you **cannot average the averages**. Multiply each group's mean by its size, add those up, then divide by the total number of items.

two classes, two different sizes

| class mean | students | mean × students |
|---|---|---|
| 75 | 20 | 1500 |
| 85 | 30 | 2550 |

**mean = 4050 ÷ 50 = 81**

The bigger class contributes more rows to the total, so the combined mean of 81 sits nearer 85 than the halfway 80.

**Example 1. A class of 20 scores a mean of 75; another class of 30 scores a mean of 85. Find the mean of all 50.**

**Solution.**
(20 × 75 + 30 × 85)/50 = (1,500 + 2,550)/50 = **81**.
Not 80 — the bigger class pulls the mean towards 85.

the 30-student class carries more weight

| class mean | students | mean × students |
|---|---|---|
| 75 | 20 | 1500 |
| 85 | 30 | 2550 |

**mean = 4050 ÷ 50 = 81**

**Example 2. A student scores 80 on a test worth 40% and 90 on one worth 60%. Find the overall score.**

**Solution.**
0.4(80) + 0.6(90) = 32 + 54 = **86**.

the weights are the percentages, and they add to 1

| score | weight | score × weight |
|---|---|---|
| 80 | 0.4 | 32 |
| 90 | 0.6 | 54 |

**mean = 86 ÷ 1 = 86**

**Example 3. 5 items cost $4 each and 3 cost $12 each. Find the mean cost.**

**Solution.**
(5 × 4 + 3 × 12)/8 = (20 + 36)/8 = **$7**.

total cost over total count

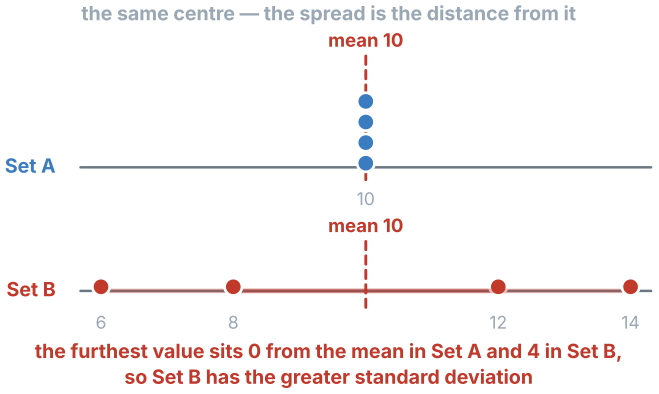
| price | how many | price × how many |
|---|---|---|
| 4 | 5 | 20 |
| 12 | 3 | 36 |

**mean = 56 ÷ 8 = 7**

## 4 Spread: range and standard deviation

**range = largest − smallest | standard deviation measures how far values sit from the mean — compare it, never compute it**

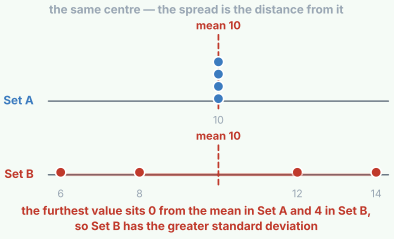
The SAT **never** asks you to calculate a standard deviation. It asks which of two data sets has the larger one, and the answer is whichever is **more spread out from its mean**. Values bunched together give a small standard deviation even if the mean is large.




Both sets centre on 10. Set A has no spread at all, so its standard deviation is zero — any spread beats none.

**Example 1.** **Set A is 10, 10, 10, 10 and Set B is 6, 8, 12, 14. Both have mean 10. Which has the greater standard deviation?**

**Solution.**
Set A has every value **at** the mean, so its standard deviation is 0.
**Set B** is greater — its values sit further from 10.




**Example 2.** **Find the range of 14, 9, 22, 9, 17.**

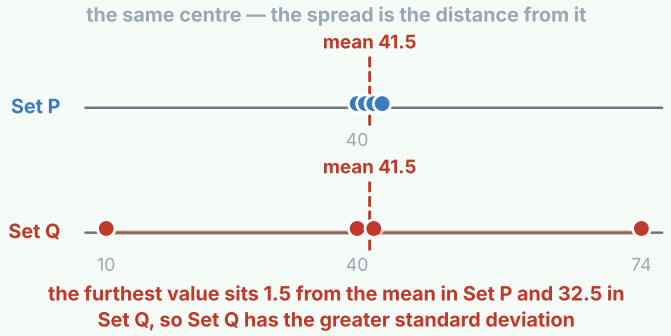
**Solution.**
22 − 9 = **13**.




the range uses only the two extremes

**Example 3.** **Set P is 40, 41, 42, 43 and Set Q is 10, 40, 42, 74. Which has the greater standard deviation, and do they have the same mean?**

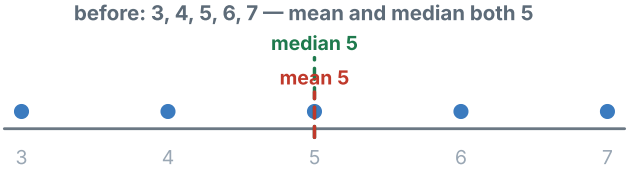
**Solution.**
Both have mean 41.5. **Set Q** has the greater standard deviation — the same centre, far wider spread.




## 5 Outliers: the mean moves, the median does not

**an extreme value drags the mean towards it and leaves the median almost alone**

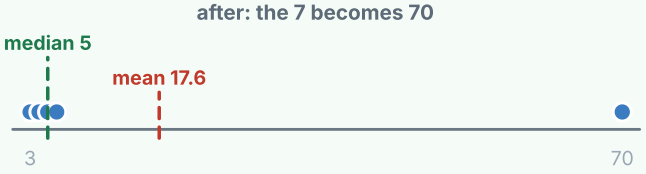
The median only cares about **position**, so one wild value changes it by at most one place. The mean adds every value into a total, so a single outlier can shift it a long way. That is why the median is the better summary of a skewed set.




This is the set before the outlier arrives. Compare it with the example below, where one value moves and only the mean follows.

**Example 1.** **For 3, 4, 5, 6, 7 the mean and median are both 5. The 7 is changed to 70. What happens to each?**

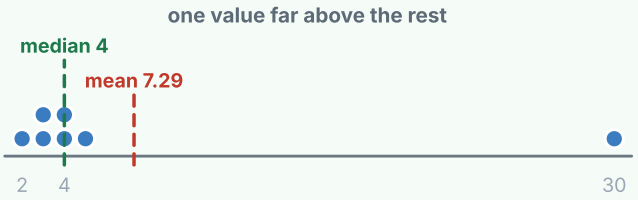
**Solution.**
New values 3, 4, 5, 6, 70: mean = 88/5 = **17.6**, median is still **5**.
The mean moved by 12.6; the median did not move at all.



**Example 2.** **A data set has an outlier far above the rest. Which is larger, the mean or the median?**

**Solution.**

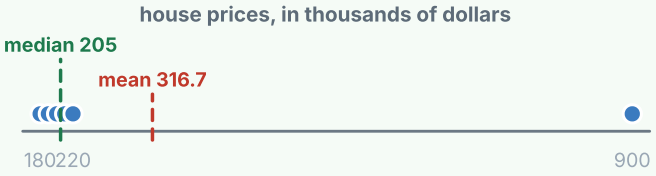
**The mean** — it is pulled up towards the outlier.




**Example 3.** **Which measure of centre should you report for house prices in a town with a few mansions?**

**Solution.**

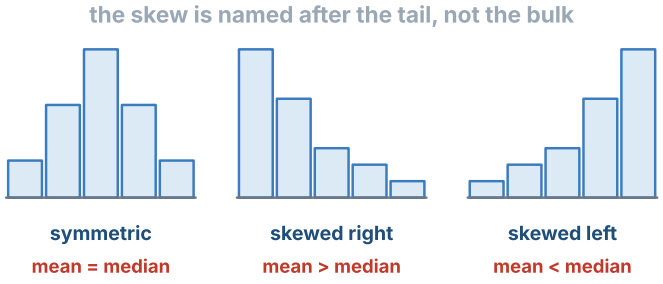
**The median.** A handful of very expensive houses would inflate the mean and misrepresent a typical house.




## 6 Shape: symmetric and skewed

**symmetric → mean = median | skewed right → mean > median | skewed left → mean < median**

The skew is named after **the tail**, not the bulk of the data. A long tail to the right drags the mean to the right of the median. Reading this off a histogram is a standard SAT question.

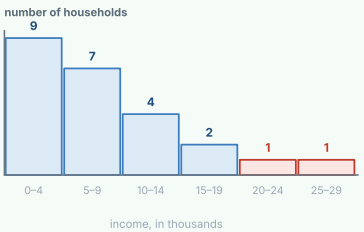



Three shapes, three verdicts. Find the tail first, then say which way the mean has been pulled.

**Example 1.** **A histogram has most of its bars on the left and a long tail stretching right. Compare the mean and the median.**

**Solution.**

Skewed right, so the tail pulls the mean up: **mean > median**.

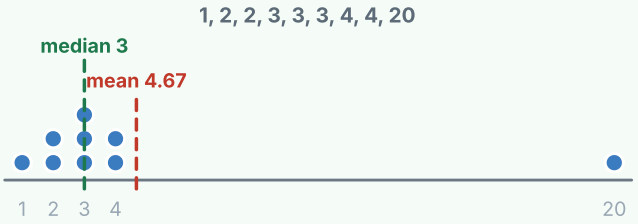


**Example 2.** **For the data 1, 2, 2, 3, 3, 3, 4, 4, 20, which is larger, the mean or the median?**

**Solution.**

Median is the 5th of 9 values = **3**. Mean = 42/9 ≈ **4.67**.

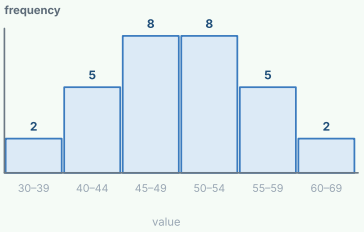
The 20 is a right tail, so mean > median. ✓




**Example 3.** **A distribution is roughly symmetric with mean 50. What is the approximate median?**

**Solution.**

**About 50.** In a symmetric distribution the two coincide.



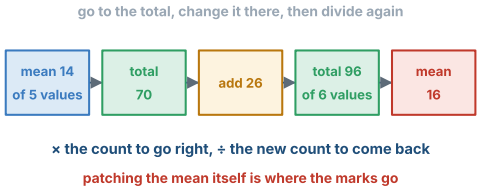
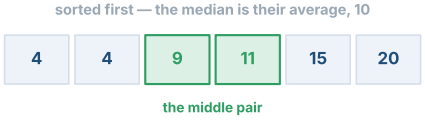
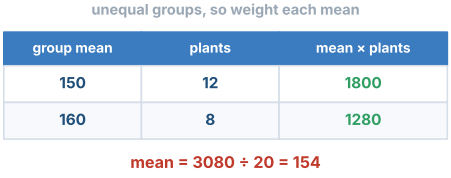
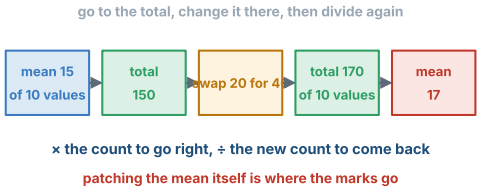
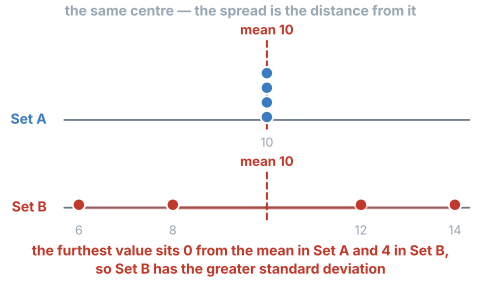
## Part 1 — With Desmos (10 questions · every one is faster on the graph)

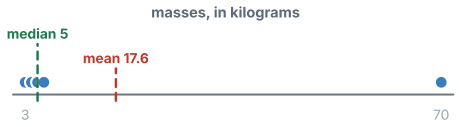
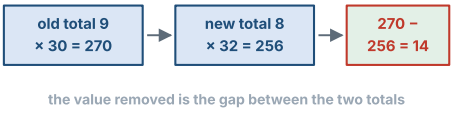
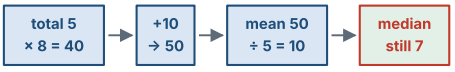
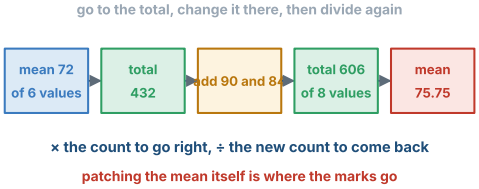
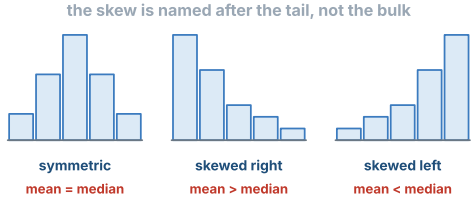
The SAT gives you Desmos on every maths question. Here it is **a calculator for totals**. The move that matters is turning a mean back into a total — total = mean × count — and you can build that straight into the line.

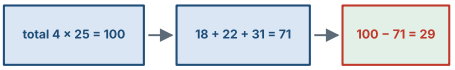
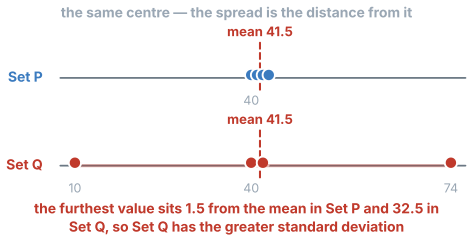
| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A basketball player scored 12, 15, 19, 22 and 27 points in five games. What is the mean number of points per game? | `(12+15+19+22+27)/5` | Type the whole sum on one line. | 19 |
| D2 | The mean mass of 6 parcels is 18 kilograms. One parcel with a mass of 23 kilograms is removed. What is the mean mass, in kilograms, of the remaining parcels? | `(6*18-23)/5` | Turn the mean into a total inside the same line. | 17 |
| D3 | Eight students have a mean test score of 12. A ninth student joins and the mean becomes 13. What is the ninth student's score? | `9*13-8*12` | New total minus old total. | 21 |
| D4 | A class of 20 students averaged 75 on a test and a class of 30 students averaged 85. What is the mean score of all 50 students? | `(20*75+30*85)/50` | Weighted — not the average of the averages. | 81 |
| D5 | The daily high temperatures, in degrees Celsius, for five days were 14, 9, 22, 9 and 17. What is the range of these temperatures? | `22-9` | Largest minus smallest. | 13 |
| D6 | Four runners finished a race in 3, 8, 5 and 12 minutes. What is the median finishing time, in minutes? | `(5+8)/2` | Sort first, then average the middle pair. | 6.5 |
| D7 | A course grade is 40% midterm and 60% final. A student scored 80 on the midterm and 90 on the final. What is the student's course grade? | `0.4*80+0.6*90` | Weights as decimals on one line. | 86 |
| D8 | The ages, in years, of five club members are 3, 4, 5, 6 and 70. What is the mean age, in years, of the members? | `(3+4+5+6+70)/5` | Compare with the median of 5 — the outlier moves only the mean. | 17.6 |
| D9 | A librarian recorded the number of books read last month by nine students: 1, 2, 2, 3, 3, 3, 4, 4 and 20. What is the mean number of books read? | `(1+2+2+3+3+3+4+4+20)/9` | Desmos keeps the exact fraction 14/3 if you want it. | 4.67 (42/9) |
| D10 | A shop sold 5 notebooks at $4 each and 3 notebooks at $12 each. What is the mean price, in dollars, per notebook sold? | `(5*4+3*12)/8` | Total cost over total count. | 7 |

## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Almost every one is solved by converting to a **total** first, adjusting it, then dividing by the **new** count — decide what the count is before you divide.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | A list of 5 numbers has a mean of 14. A sixth number, 26, is added to the list. What is the mean of the 6 numbers?<br>  | 16 | Total of the five is 5 × 14 = 70. New total 70 + 26 = 96 over 6 numbers.<br>96 ÷ 6 = **16**. The trap for forgetting the sixth value. |
| P2 | Hard | The number of hours worked last week by each of 6 employees was 11, 4, 9, 4, 15 and 20. What is the median number of hours worked?<br>  | 10 | Sorted: 4, 4, 9, 11, 15, 20. Even count, so average the middle pair.<br>(9 + 11)/2 = **10**. A and D are the two middle values taken alone. |
| P3 | Hard | In an experiment, 12 plants had a mean height of 150 centimetres and 8 plants had a mean height of 160 centimetres. What is the mean height, in centimetres, of all 20 plants?<br>  | 154 | (12 × 150 + 8 × 160)/20 = 3,080/20 = **154**.<br>The trap: averaging 150 and 160 ignores the unequal group sizes. |
| P4 | Hard | A list of 10 numbers has a mean of 15. One of the numbers, 20, is replaced by 40 and no other number is changed. What is the mean of the new list?<br>  | 17 | The total rises by 40 − 20 = 20, from 150 to 170.<br>170 ÷ 10 = **17**. |
| P5 | Hard | Data set A consists of the values 10, 10, 10 and 10. Data set B consists of the values 6, 8, 12 and 14. State what is true here.<br>  | Data set B has the greater standard deviation. | Both sets have a mean of 10, but every value of A **is** the mean, so A has no spread at all and its standard deviation is 0.<br>Any spread beats none. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P6 | Hard | The masses, in kilograms, of 5 packages are 3, 4, 5, 6 and 70. Which correctly compares the mean and the median of these masses?<br>  | **The mean is greater than the median.** | Mean = 88/5 = **17.6**; median = **5**.<br>The single large value pulls the mean up but leaves the middle value where it was. |
| P7 | Very hard | The mean of a list of 9 numbers is 30. When one number is removed from the list, the mean of the remaining 8 numbers is 32. What is the value of the number that was removed?<br>  <br>the value removed is the gap between the two totals | **14** | Old total 9(30) = 270; the remaining 8 total 8(32) = 256.<br>270 − 256 = **14**. Removing a below-average value raises the mean. |
| P8 | Very hard | A list of 5 numbers has a mean of 8 and a median of 7. The largest number in the list is increased by 10, and no other number is changed. What are the mean and the median of the new list?<br>  <br>the biggest value is still biggest, so no one changes places | **mean 10, median 7** | Total 5(8) = 40 → 50, so the mean becomes 50/5 = **10**.<br>The largest value is still the largest, so the sorted order is unchanged and the middle value — the median — stays **7**. |
| P9 | Very hard | Six students in a class had a mean test score of 72. Two more students, with scores of 90 and 84, join the class. What is the mean score of all 8 students?<br>  | **75.75** | Old total 6(72) = 432; new total 432 + 90 + 84 = 606 over 8 students.<br>606 ÷ 8 = **75.75**. |
| P10 | Very hard | A histogram showing the distribution of household incomes in a city is skewed to the right. Which statements about the mean and the median income is true?<br>  | **The mean is greater than the median.** | The long right tail contains a few very large incomes.<br>They enter the mean at full weight but shift the middle value hardly at all, so the mean is dragged **above** the median. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P11 | Very hard | The mean of 4 numbers is 25. Three of the numbers are 18, 22 and 31. What is the value of the fourth number?<br>  | 29 | Total = 4 × 25 = 100. Known three sum to 71.<br>100 − 71 = **29**. The trap for stopping at the sum of the three. |
| P12 | Very hard | Data set P consists of 40, 41, 42 and 43. Data set Q consists of 10, 40, 42 and 74. The two data sets have the same mean. State what is true here.<br>  | Data set Q has the greater standard deviation. | Both have mean 166/4 = 41.5. P's values all sit within 1.5 of the centre; Q's reach 31.5 away.<br>Standard deviation measures typical distance from the mean, so **Q's is far larger**. |

**The one habit to build.** Never patch a mean directly. Turn it into a **total**, change the total, then divide by the **new** number of values. Half of these twelve change the count as well as the total, and that is where the marks go.

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*

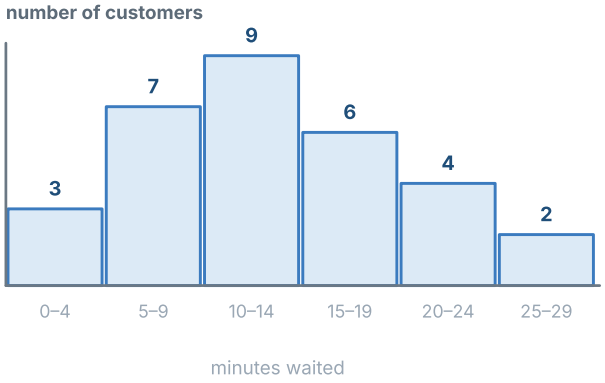
# SAT Math 3.4 — Data Displays

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Reading a bar chart or histogram

**a bar chart compares separate categories | a histogram shows how one number is distributed across intervals**

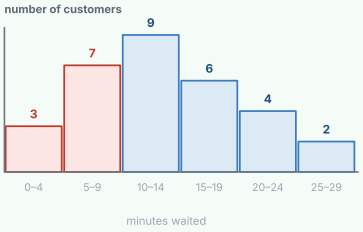
**Read the scale before the bars.** A bar chart has gaps between the bars because the categories are separate things; a histogram has none, because the intervals run into each other. The height is a **count**, so the total sample size is the sum of all the heights — not the number of bars.




Six touching intervals. The sample size is 3 + 7 + 9 + 6 + 4 + 2 = 31 customers, and six is only the number of bars.

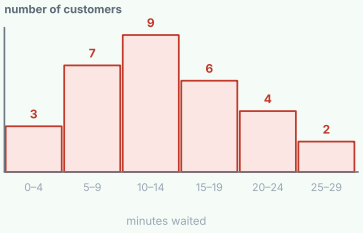
**Example 1. Using the histogram above, how many customers waited fewer than 10 minutes?**

**Solution.**
The first two intervals are 0–4 and 5–9, with heights 3 and 7.
3 + 7 = **10 customers**.




**Example 2. How many customers are in the sample altogether?**

**Solution.**
Add every bar: 3 + 7 + 9 + 6 + 4 + 2 = **31**. A common slip is to answer 6, the number of bars.



**Example 3.** **Which interval contains the median wait?**

**Solution.**
With 31 values the median is the 16th. Running totals: 3, 10, 19 — the 16th falls in the third interval, **10–14 minutes**.




| ning total | 3 | 10 | 19 | 25 | 29 | 31 |
|---|---|---|---|---|---|---|

31 values in all, so the middle one is the 16th — the running total first reaches it in 10–14

## 2 Frequency tables and the mean

**mean from a table = ∑(value × frequency) ÷ ∑frequency — divide by the total count, never by the number of rows**

This is the **single most common error** on data-display questions. A table with five rows does not mean you divide by five. Multiply each value by how many times it occurred, add those products, then divide by the **total frequency**.

three rows, but twenty values

| value | frequency | value × freq |
|---|---|---|
| 1 | 4 | 4 |
| 2 | 6 | 12 |
| 3 | 10 | 30 |

**mean = 46 ÷ 20 = 2.3**

The right-hand column is what you add; the middle column is what you divide by. The number of rows plays no part at all.

**Example 1.** **Values 1, 2, 3 occur 4, 6 and 10 times. What is the mean?**

**Solution.**
∑(value × freq) = 1(4) + 2(6) + 3(10) = 4 + 12 + 30 = 46.
Total frequency = 20, so the mean is 46/20 = **2.3**.

46 over 20, not 46 over 3

| value | frequency | value × freq |
|---|---|---|
| 1 | 4 | 4 |
| 2 | 6 | 12 |
| 3 | 10 | 30 |

**mean = 46 ÷ 20 = 2.3**

**Example 2.** **A quiz is marked out of 4. Scores 2, 3 and 4 were earned by 5, 8 and 7 students. Find the mean.**

**Solution.**

2(5) + 3(8) + 4(7) = 10 + 24 + 28 = 62 over 20 students.

62/20 = **3.1**.

a quiz marked out of 4

| score | students | score × students |
|---|---|---|
| 2 | 5 | 10 |
| 3 | 8 | 24 |
| 4 | 7 | 28 |

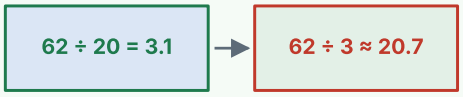
mean = 62 ÷ 20 = 3.1

**Example 3.** **Why is dividing by 3 wrong in the question above?**

**Solution.**

Three is the number of **distinct scores**, not the number of students.

62/3 ≈ 20.7, which is not even a possible score out of 4 — a quick sanity check that catches the error.

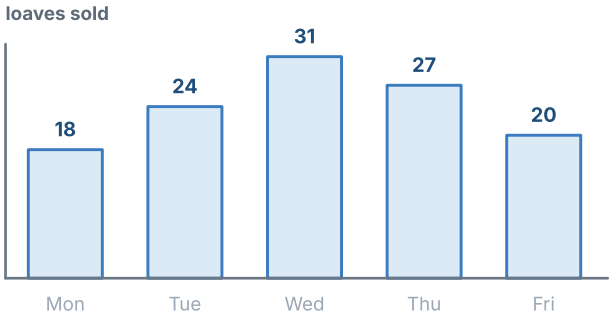



a score of 20.7 out of 4 is impossible — check the divisor

## 3 Bar charts: comparing categories

**read the axis scale first | ‘how many more’ means subtract, ‘how many times’ means divide**

Most bar-chart questions are one of three things: **read one bar, compare two bars, or combine several.** The wording tells you which operation. Watch for an axis that does not start at zero — it makes differences look bigger than they are.

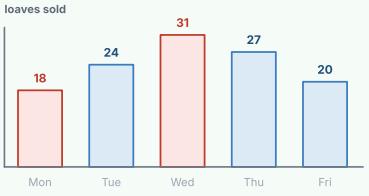



separate days, so the bars stand apart

Five separate days. Nothing here is an interval, so nothing touches — that gap is what tells you it is a bar chart.

**Example 1.** **From the chart above, how many more loaves were sold on Wednesday than on Monday?**

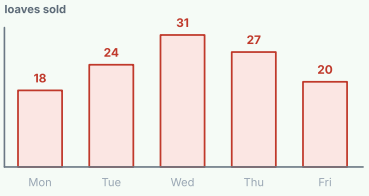
**Solution.**
Wednesday 31, Monday 18.
31 − 18 = **13 loaves**.




‘how many more’ — subtract the two

**Example 2.** **What was the total number of loaves sold over the five days?**

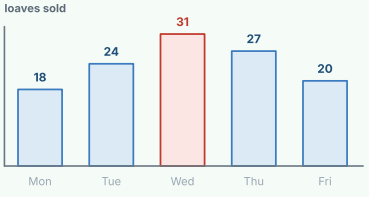
**Solution.**
18 + 24 + 31 + 27 + 20 = **120 loaves**.




every bar counts towards the week’s total of 120

**Example 3.** **What percentage of the week's sales happened on Wednesday?**

**Solution.**
31 out of 120.
31/120 = 0.2583… ≈ **25.8%**.




one bar over the total of all five: 31 ÷ 120

## 4 Two-way tables

**each cell is a count | the row and column totals must add to the grand total**

Fill in any missing totals **before** you answer anything. Then the only question that matters is **which number is the whole**: the grand total, one row total, or one column total. The wording picks it for you.

Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

Three possible denominators sit in this one table: 180, a row total of 100 or 80, or a column total of 80 or 100.

**Example 1. From the table above, how many customers bought cheese?**

**Solution.**

The Cheese column totals 60 + 20 = **80 customers**.

Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

**Example 2. What fraction of all customers bought fruit but not cheese?**

**Solution.**

That cell holds 40, and the grand total is 180.

40/180 = **2/9 ≈ 0.222**.

Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

**Example 3.** **Of the customers who bought cheese, what fraction also bought fruit?**

**Solution.**

Now the whole is the Cheese column, **80** — not 180.

60/80 = **3/4**. Changing which total you divide by is the whole skill here.

Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

## 5 Dot plots and the shape of data

**every dot is one value | count dots for the sample size, not columns**

A dot plot shows the **whole data set**, so you can read the mode straight off (the tallest stack), count to the median, and see the shape. Repeated values matter: three dots above 4 means the value 4 appears three times in the list.

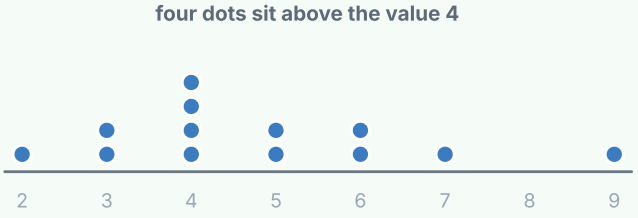



Thirteen dots, eight columns. The sample size is the number of dots, and the tallest stack is the mode.

**Example 1.** **Using the dot plot above, how many students read exactly 4 books?**

**Solution.**

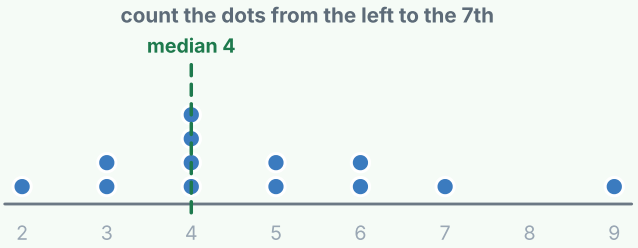
There are four dots above 4, so **4 students**.



**Example 2.** **What is the median number of books read?**

**Solution.**

13 values, so the median is the 7th. Counting up: 2 (1), 3 (3), 4 (7) — the 7th dot sits above **4 books**.




**Example 3.** **What is the range?**

**Solution.**

Largest 9, smallest 2.

9 − 2 = **7 books**.

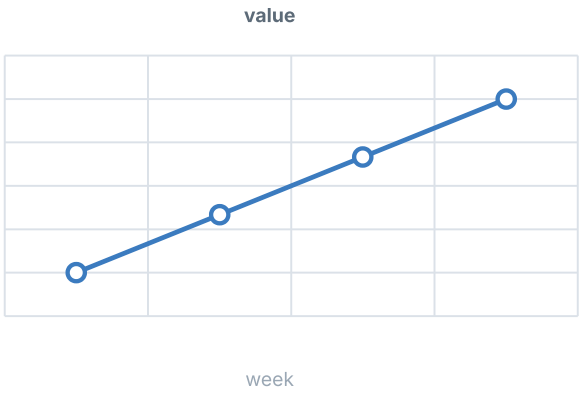



the range uses only the two ends of the plot

## 6 Line graphs and rates of change

**steeper = faster | flat = no change | downward = decreasing**

On a line graph the question is almost always about the **slope** between two points, not the heights themselves. ‘Greatest increase’ means the **steepest** upward segment, which is not necessarily the segment ending highest.




Equal steps of 4 a week give a constant slope. The rate of change is (32 − 20) ÷ (4 − 1) = 4 per week.

**Example 1.** **A graph shows 20 at week 1 and 32 at week 4. What is the average rate of change per week?**

**Solution.**
(32 − 20)/(4 − 1) = 12/3 = **4 per week**.



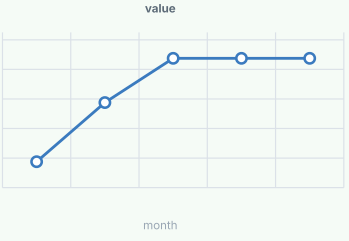

**Example 2.** **Between which pair of days did sales rise fastest: Mon→Tue (18 to 24) or Tue→Wed (24 to 31)?**

**Solution.**
Mon→Tue rises 6; Tue→Wed rises 7.
The faster rise is **Tuesday to Wednesday**.




**Example 3.** **A line graph is flat from month 3 to month 5. What does that mean?**

**Solution.**
The quantity **did not change** over those two months.
Flat does not mean zero — it means constant, and the constant could be any height.



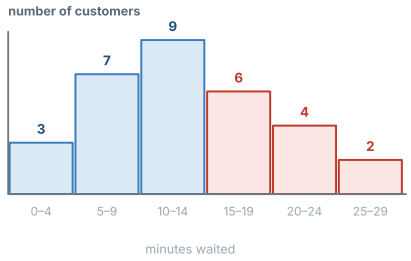
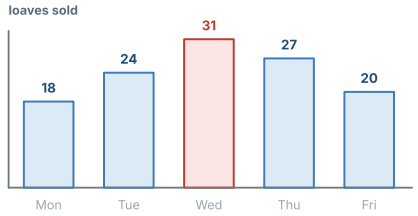
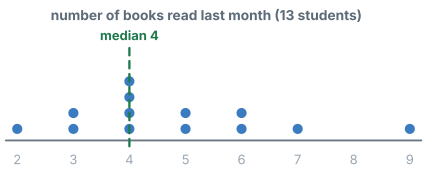
## Part 1 — With Desmos (10 questions · every one is faster on the graph)

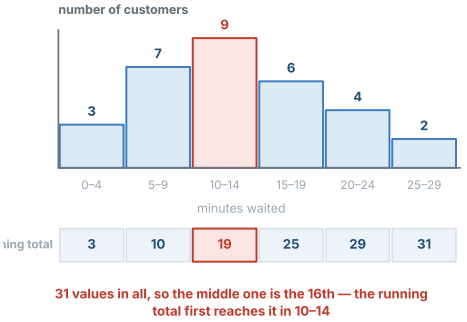
The SAT gives you Desmos on every maths question. Here it is **a calculator plus a careful denominator**. Build the frequency products and the correct total into a single line so you cannot divide by the wrong thing.

| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A shop's daily sales were 18, 24, 31, 27 and 20 loaves. What is the mean number of loaves sold per day? | `(18+24+31+27+20)/5` | Type the whole sum on one line. | 24 |
| D2 | In a survey the values 1, 2 and 3 occurred 4, 6 and 10 times. What is the mean value? | `(1*4+2*6+3*10)/(4+6+10)` | Products on top, total frequency underneath. | 2.3 |
| D3 | A histogram has bars of height 3, 7, 9, 6, 4 and 2. How many values are in the data set? | `3+7+9+6+4+2` | Add the heights, not the bars. | 31 |
| D4 | Of 180 customers surveyed, 100 bought fruit and 60 of those also bought cheese. What fraction of the fruit buyers bought cheese? | `60/100` | The fruit buyers are the whole here. | 0.6 |
| D5 | A quiz out of 4 was scored 2, 3 and 4 by 5, 8 and 7 students. What is the mean score? | `(2*5+3*8+4*7)/20` | Weighted mean in one line. | 3.1 |
| D6 | A bakery sold 31 loaves on Wednesday out of 120 for the week. What percentage of the week's loaves is that, to the nearest tenth? | `31/120*100` | Desmos gives the full decimal; round at the end. | 25.8 |
| D7 | A line graph reads 20 at week 1 and 32 at week 4. What is the average rate of change per week? | `(32-20)/(4-1)` | Rise over run. | 4 |
| D8 | A dot plot shows 2, 3, 3, 4, 4, 4, 4, 5, 5, 6, 6, 7 and 9. What is the mean, to two decimal places? | `(2+3+3+4+4+4+4+5+5+6+6+7+9)/13` | Desmos holds the exact fraction 62/13. | 4.77 |
| D9 | A two-way table has row totals 100 and 80. What percentage of the whole sample is the first row? | `100/180*100` | Round only at the end. | 55.6 |
| D10 | A histogram's intervals 0–4 and 5–9 hold 3 and 7 values out of 31. What percentage of the data is below 10? | `10/31*100` | Combine the two bars first. | 32.3 |

## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Before touching the numbers, decide what the display **shows** and what the question **wants**: a value, a difference, a total, or a fraction — and if a fraction, out of what.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | The histogram shows the time, in minutes, that each of 31 customers waited at a service desk. How many customers waited 15 minutes or longer?<br>  | 12 | The last three intervals are 15–19, 20–24 and 25–29, with heights 6, 4 and 2.<br>6 + 4 + 2 = **12**. A common wrong answer is the first of those bars taken alone. |
| P2 | Hard | In a frequency table, the value 1 occurs 4 times, the value 2 occurs 6 times and the value 3 occurs 10 times. What is the mean of the data?<br>divide by 20, not by 3 and not by 7<br>value \| frequency \| value × freq<br>1 \| 4 \| 4<br>2 \| 6 \| 12<br>3 \| 10 \| 30<br>mean = 46 ÷ 20 = 2.3 | 2.3 | ∑(value × freq) = 4 + 12 + 30 = 46 over a total frequency of 20.<br>46/20 = **2.3**. A common wrong answer is the mean of 1, 2 and 3 with the frequencies ignored; D is 46 ÷ 7. |
| P3 | Hard | The table shows the results of a survey of 180 market customers. Of the customers who bought cheese, what fraction also bought fruit?<br>Survey of 180 market customers<br>\| \| Cheese \| No cheese \| Total \|<br>Bought fruit \| 60 \| 40 \| 100<br>No fruit \| 20 \| 60 \| 80<br>Total \| 80 \| 100 \| 180 | 3/4 | The whole is the Cheese column, 60 + 20 = 80, not the grand total.<br>60/80 = **3/4**. A common wrong answer divides by the Fruit row (100) instead. |
| P4 | Hard | The bar chart shows loaves sold on each of five days. What percentage of the week's total was sold on Wednesday, to the nearest tenth of a percent?<br>  | 25.8 | Total = 18 + 24 + 31 + 27 + 20 = 120.<br>31/120 = 0.2583… = **25.8%**. |
| P5 | Hard | The dot plot shows the number of books read last month by each of 13 students. What is the median number of books read?<br>  | 4 | Thirteen values, so the median is the 7th in order. The running count 2 (1), 3 (3), 4 (7) puts the 7th value at **4**.<br>A common wrong answer is the mean, which the question did not ask for. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P6 | Hard | A bakery's sales rose from 18 loaves on Monday to 24 on Tuesday, then to 31 on Wednesday. Between which pair of consecutive days was the increase greatest, and by how much?<br>  | **Tuesday to Wednesday, by 7** | Monday to Tuesday rises 24 − 18 = 6; Tuesday to Wednesday rises 31 − 24 = 7. The greater rise is **Tuesday to Wednesday, by 7**. B and C give day totals, not increases. |
| P7 | Very hard | In the two-way table, what percentage of the customers who did not buy fruit also did not buy cheese?<br>Survey of 180 market customers<br>\| | Cheese | No cheese | Total \|<br>\| Bought fruit \| 60 \| 40 \| 100 \|<br>\| No fruit \| 20 \| 60 \| 80 \|<br>\| Total \| 80 \| 100 \| 180 \| | **75** | The whole is the 'No fruit' row, which totals 20 + 60 = 80.<br>60/80 = **75%**. Dividing 60 by the grand total 180 gives 33.3%, which answers a different question. |
| P8 | Very hard | A histogram of 31 wait times has bars of height 3, 7, 9, 6, 4 and 2 over the intervals 0–4, 5–9, 10–14, 15–19, 20–24 and 25–29 minutes. Which interval contains the median wait time?<br>  | **10–14** | With 31 values the median is the 16th. Running totals are 3, 10, 19, so the 16th value sits in the third interval, **10–14**.<br>A common wrong answer stops at the running total 10. |
| P9 | Very hard | A dot plot shows the values 2, 3, 3, 4, 4, 4, 4, 5, 5, 6, 6, 7 and 9. One more student is added who read 9 books. What happens to the median?<br>  | **It rises to 4.5.** | With 14 values the median is the average of the 7th and 8th. In order those are 4 and 5. (4 + 5)/2 = **4.5**. Adding a large value at the top shifts the middle by half a step, not a whole one. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P10 | Very hard | A frequency table records a quiz marked out of 4: five students scored 2, eight scored 3 and seven scored 4. A student computes the mean as 62 ÷ 3 ≈ 20.7. What is the error, and what is the correct mean?<br>twenty students, three rows<br>score \| students \| score × students<br>2 \| 5 \| 10<br>3 \| 8 \| 24<br>4 \| 7 \| 28<br>mean = 62 ÷ 20 = 3.1 | **Divided by the number of distinct scores instead of the number of students; the mean is 3.1.** | ∑(value × freq) = 10 + 24 + 28 = 62 is right, but the divisor must be the total frequency, **20**, not the 3 rows.<br>62/20 = **3.1**. A mean of 20.7 on a test marked out of 4 is impossible, which is the quickest way to spot it. |
| P11 | Very hard | Over five days a bakery sold 18, 24, 31, 27 and 20 loaves. On the sixth day it sold enough to raise the mean for all six days to 25. How many loaves were sold on the sixth day?<br>five days total 120 → six-day total 6 × 25 = 150 → 150 − 120 = 30<br>work in totals, then take the difference | **30** | Five-day total = 120. For a six-day mean of 25 the total must be 6 × 25 = 150.<br>150 − 120 = **30**. |
| P12 | Very hard | In a two-way table of 180 customers, 100 bought fruit and 80 did not; 60 fruit buyers and 20 non-buyers bought cheese. A report claims ‘most cheese buyers did not buy fruit’. Which statement best evaluates the claim?<br>Survey of 180 market customers<br>\| \| Cheese \| No cheese \| Total<br>Bought fruit \| 60 \| 40 \| 100<br>No fruit \| 20 \| 60 \| 80<br>Total \| 80 \| 100 \| 180 | **It is incorrect: 60 of the 80 cheese buyers also bought fruit.** | There are 80 cheese buyers in all, and 60 of them bought fruit — **three quarters**.<br>The claim reverses the comparison. A common wrong answer quotes a true number that answers a different question. |

**The one habit to build.** In this topic the answer is almost always right and **the denominator is almost always wrong**. Grand total, row total, column total, total frequency — name it out loud before you divide.

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*

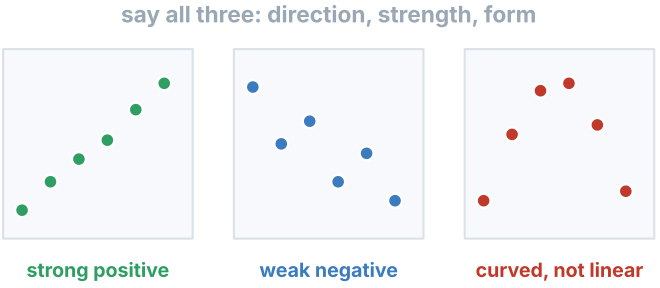
# SAT Math 3.5 — Two-Variable Data & Models

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Describing a scatterplot

**say three things: direction (up or down) | strength (tight or scattered) | form (straight or curved)**

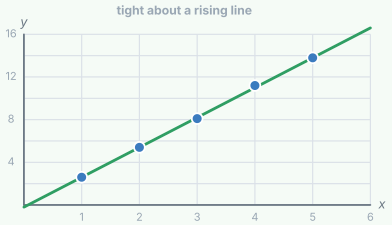
**Direction** is whether y rises or falls as x rises. **Strength** is how closely the points hug a line. **Form** is whether that pattern is a line at all — a tight parabola is a strong association but not a linear one. The SAT asks you to name all three, and a strong curve is the classic trap.




Three independent judgements. The third panel is strong *and* not linear, which is exactly the combination the test likes.

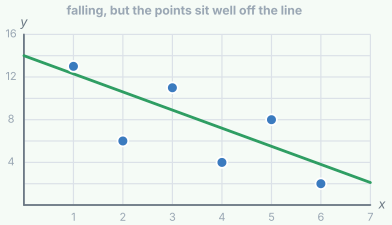
**Example 1.** **Points rise from bottom-left to top-right and lie very close to a line. Describe the association.**

**Solution.**
**Strong positive linear association.** Rising is positive, hugging the line is strong, straight is linear.



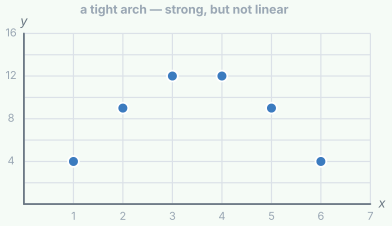

**Example 2.** **Points fall as x rises but are widely scattered. Describe it.**

**Solution.**
**Weak negative linear association.** Falling is negative; the scatter makes it weak.




**Example 3.** **Points rise, level off, then fall, forming a tight arch. Is this a strong linear association?**

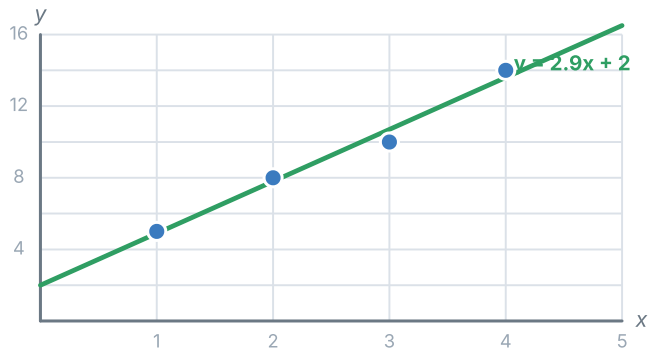
**Solution.**
**No.** It is a strong association but not a linear one.
Strength and form are separate judgements, and a line of best fit would describe this badly.



## 2 The line of best fit

**the line of best fit minimises the total squared vertical distance | in Desmos type $y_1 \sim m x_1 + b$ under a table**

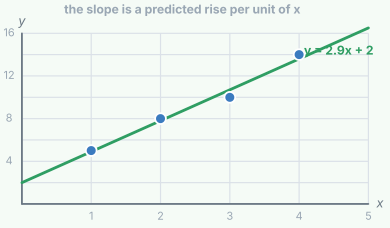
The line **does not have to pass through any data point**, and usually passes through none. Its **slope** is the predicted change in y for a one-unit rise in x, and its **y-intercept** is the predicted y when x is zero — which may be meaningless in context.




Four points and the line Desmos reports. It misses every one of them, which is normal and not an error.

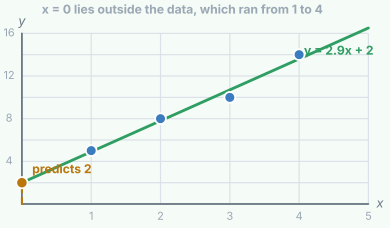
**Example 1.** **For the data (1, 5), (2, 8), (3, 10), (4, 14), Desmos reports y = 2.9x + 2. What does 2.9 mean?**

**Solution.**
For every increase of 1 in x, the model predicts y rises by about **2.9**.



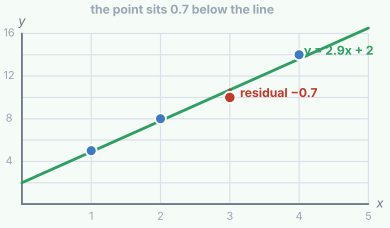

**Example 2.** **What does the intercept 2 mean in that model?**

**Solution.**
It is the predicted value of y when x = 0, so **about 2**.
If x = 0 is outside the data range, this is extrapolation and may have no sensible meaning.




**Example 3.** **Does the line pass through (3, 10)?**

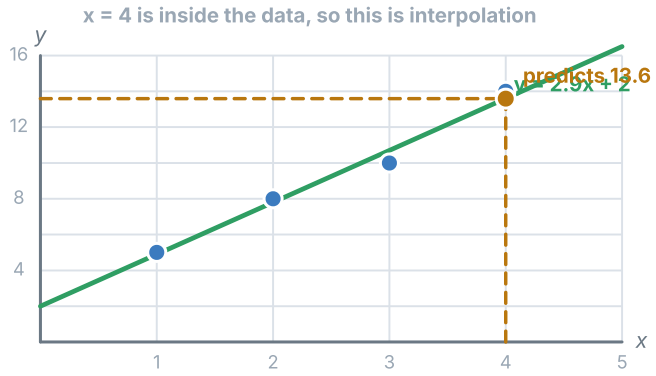
**Solution.**
At x = 3 the line gives 2.9(3) + 2 = **10.7**, not 10.
**No** — a line of best fit need not pass through any data point.



## 3 Predicting from a model

**substitute the x you are given | inside the data range is interpolation, outside is extrapolation**

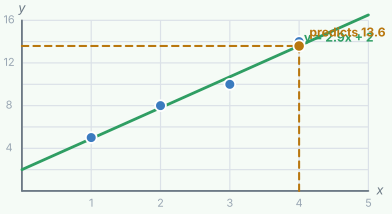
Prediction is just substitution. The judgement the SAT tests is whether the prediction is **trustworthy**: inside the range of the data it usually is; far outside it, the model has no evidence behind it and the prediction is unreliable however neat the arithmetic.




Read up from the x you are given, across to the y the line gives. Whether to trust it depends only on where x sits.

**Example 1.** **Using y = 2.9x + 2, predict y when x = 4.**

**Solution.**
2.9(4) + 2 = 11.6 + 2 = **13.6**.




**Example 2.** **The data covers x from 1 to 4. How much should you trust a prediction at x = 30?**

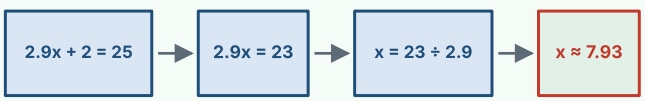
**Solution.**
**Very little.** x = 30 is far outside the data, so this is **extrapolation** — the model has no evidence that the pattern continues that far.

the arithmetic is fine; the evidence is not

| | |
|---|---|
| x between 1 and 4 | interpolation — supported by the data |
| x = 30 | extrapolation — no evidence at all |

**Example 3.** **Using y = 2.9x + 2, for what x does the model predict y = 25?**

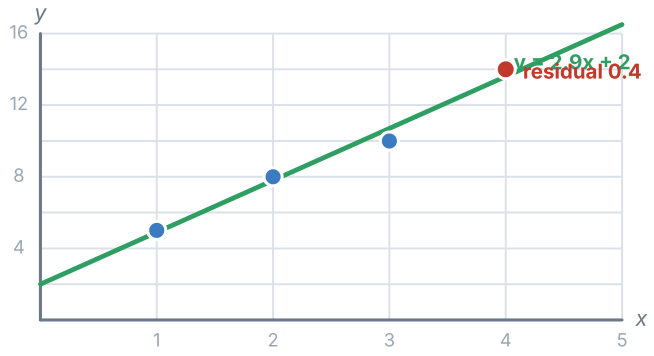
**Solution.**
2.9x + 2 = 25, so 2.9x = 23 and x = 23/2.9 ≈ **7.93**.



## 4 Residuals

**residual = actual − predicted | positive means the point sits above the line**

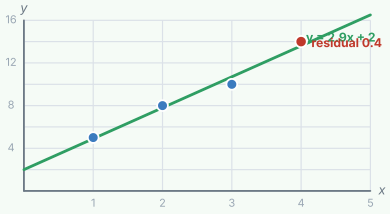
A residual measures **one point's miss**. Subtract in that order every time: **actual first**. A residual plot with no pattern says a line was the right model; a residual plot with a curve says it was not.




The red segment is the residual: the vertical gap from the line up to the actual point, +0.4 here.

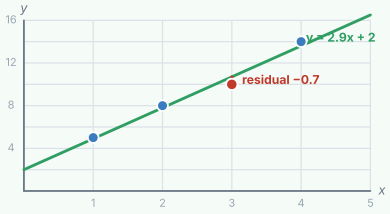
**Example 1.** **The model predicts 13.6 and the actual value is 14. Find the residual.**

**Solution.**
14 − 13.6 = **+0.4**. Positive, so the point lies just **above** the line.



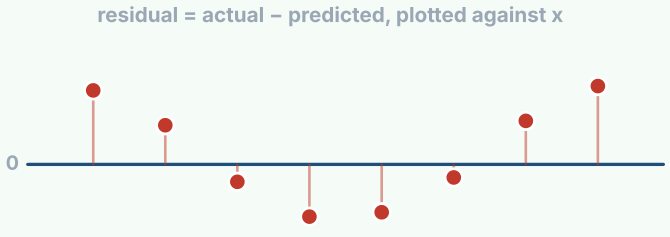

**Example 2.** **The model predicts 10.7 and the actual value is 10. Find the residual.**

**Solution.**
10 − 10.7 = **−0.7**. Negative, so this point lies **below** the line.




**Example 3.** **A residual plot shows a clear U shape. What does that tell you?**

**Solution.**
The straight-line model is **not appropriate** — the data curves, and a line systematically over- then under-predicts.



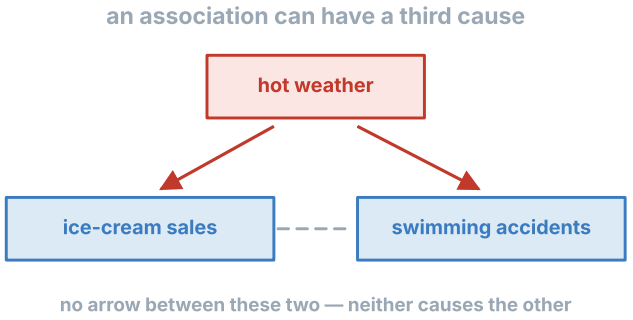

**a clear U shape**

the data curves — a straight line is the wrong model

## 5 Correlation is not causation

**an association shows two things move together | only a randomised experiment shows one causes the other**

This is tested **in words, not numbers**. However strong the association, an observational study cannot establish cause. Watch for a **lurking third variable** that drives both quantities.



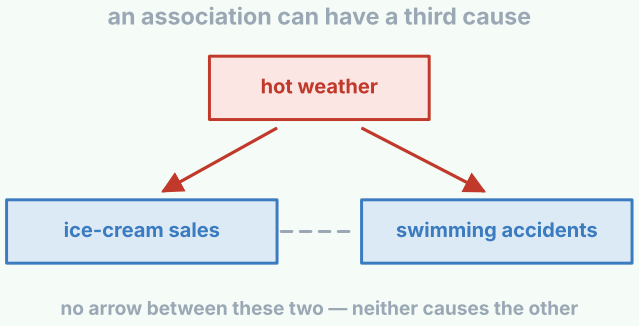

Two arrows out of one cause, and no arrow between the two effects. That missing arrow is the whole point.

**Example 1.** **Ice-cream sales and swimming accidents rise together. Does ice cream cause accidents?**

**Solution.**

**No.** Hot weather raises both — a lurking variable.

The association is real; the causal claim is not.

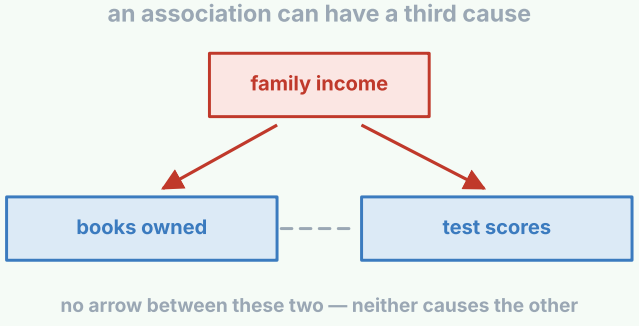



**Example 2.** **A study finds students who own more books score higher. Can the school conclude books raise scores?**

**Solution.**

**No.** This is observational, so it shows association only.

Family income could drive both.



**Example 3.** **What would be needed to support a causal claim?**

**Solution.**
**A randomised controlled experiment**: assign students to conditions at random, so the groups differ only in the treatment.

only one design licenses the word ‘causes’

| | |
|---|---|
| observational study | association only |
| random assignment | cause can be claimed |

## 6 Exponential against linear growth

**constant difference → linear | constant ratio → exponential**

Check the table before choosing a model. **Subtract** consecutive y-values: if those differences are about equal, the model is linear. **Divide** consecutive y-values instead: if those ratios are about equal, it is exponential. Exponential growth eventually overtakes any linear growth.

| | x = 1 | x = 2 | x = 3 | x = 4 |
|---|---|---|---|---|
| | 3 | 6 | 12 | 24 |
| differences | | +3 | +6 | +12 |
| ratios | | ×2 | ×2 | ×2 |

equal ratios → exponential

Test the table both ways every time. Here the differences grow but the ratios hold at 2, so the model is exponential.

**Example 1.** **y goes 3, 7, 11, 15 as x goes 1, 2, 3, 4. Which model fits?**

**Solution.**
Differences 4, 4, 4 are constant, so the model is **linear, y = 4x − 1**.

| | x = 1 | x = 2 | x = 3 | x = 4 |
|---|---|---|---|---|
| | 3 | 7 | 11 | 15 |
| differences | | +4 | +4 | +4 |
| ratios | | ×2.33 | ×1.57 | ×1.36 |

equal differences → linear

**Example 2.** **y goes 3, 6, 12, 24. Which model fits?**

**Solution.**

Ratios 2, 2, 2 are constant, so the model is **exponential, $y = 1.5(2)^x$**.

| | x = 1 | x = 2 | x = 3 | x = 4 |
|---|---|---|---|---|
| | 3 | 6 | 12 | 24 |
| differences | | +3 | +6 | +12 |
| ratios | | ×2 | ×2 | ×2 |

equal ratios → exponential

**Example 3.** **y goes 100, 90, 81, 72.9. Linear or exponential?**

**Solution.**

Differences −10, −9, −8.1 are **not** constant, but ratios are all 0.9.

**Exponential decay**, falling 10% each step.

| | x = 0 | x = 1 | x = 2 | x = 3 |
|---|---|---|---|---|
| | 100 | 90 | 81 | 72.9 |
| differences | | −10 | −9 | −8.1 |
| ratios | | ×0.9 | ×0.9 | ×0.9 |

equal ratios → exponential

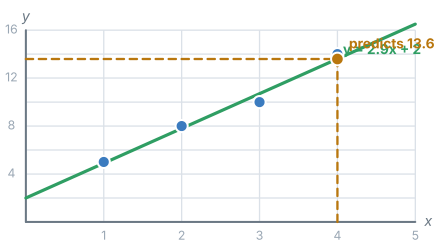
## Part 1 — With Desmos (10 questions · every one is faster on the graph)

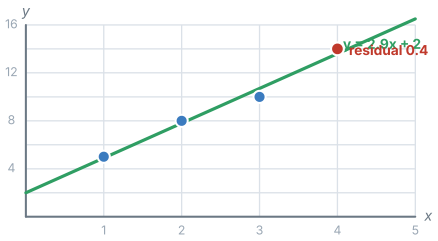
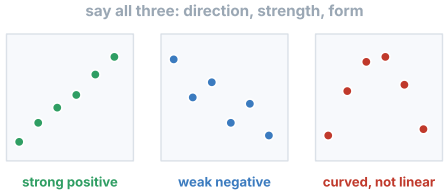
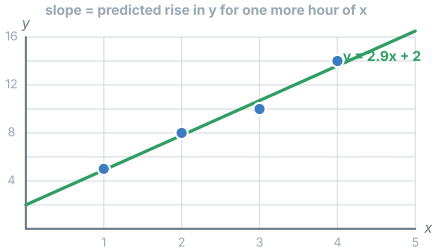
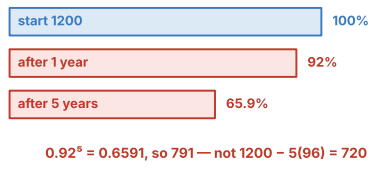
The SAT gives you Desmos on every maths question. This is the one data topic where Desmos really earns its place: **type a table and the regression line $y_1 \sim mx_1+b$**, and it reports the slope and intercept directly.

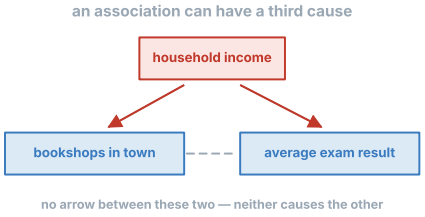
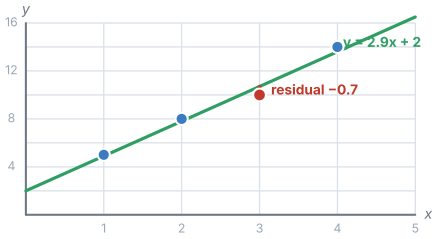
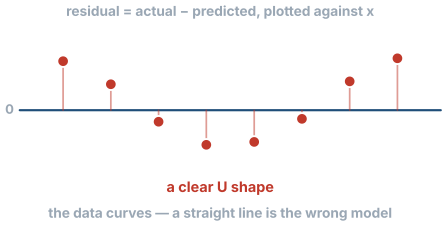
| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A table lists (1, 5), (2, 8), (3, 10) and (4, 14). What is the slope of the line of best fit, to one decimal place? | `y_1~mx_1+b` | Enter the table, then type the regression under it. | 2.9 |
| D2 | For that same data, what is the y-intercept of the line of best fit? | `y_1~mx_1+b` | Desmos reports m and b together. | 2 |
| D3 | Using y = 2.9x + 2, what value does the model predict when x = 4? | `2.9*4+2` | Substitute and let Desmos evaluate. | 13.6 |
| D4 | The actual value at x = 4 is 14 and the model predicts 13.6. What is the residual? | `14-(2.9*4+2)` | Actual minus predicted, in that order. | 0.4 |
| D5 | Using y = 2.9x + 2, for what value of x does the model predict y = 25? Round to two decimal places. | `(25-2)/2.9` | Solve the linear equation in one line. | 7.93 |
| D6 | A table shows y = 3, 6, 12, 24 for x = 1, 2, 3, 4. What is the common ratio? | `6/3` | Divide consecutive terms to test for an exponential model. | 2 |
| D7 | A population of 1200 falls by 8% each year. What is it after 5 years, to the nearest whole number? | `1200*0.92^5` | A fall of 8% means a factor of 0.92. | 791 |
| D8 | A line of best fit is y = 4x − 1. What does it predict at x = 9? | `4*9-1` | Interpolation or extrapolation depends on the data range, but the arithmetic is the same. | 35 |
| D9 | Data points are (2, 11) and (6, 23). What is the slope of the line through them? | `(23-11)/(6-2)` | Rise over run. | 3 |
| D10 | A model predicts 72.9 and the actual value is 70. What is the residual, to one decimal place? | `70-72.9` | A negative residual means the point is below the line. | −2.9 |

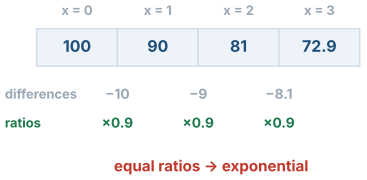
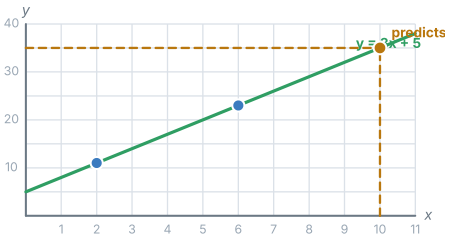
## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Three things get tested: **what the slope and intercept mean**, **how far the model may be trusted**, and **what the data does and does not let you claim**.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | The scatterplot shows four data points and the line of best fit y = 2.9x + 2. What value does the model predict when x = 4?<br>  | 13.6 | 2.9(4) + 2 = 11.6 + 2 = **13.6**.<br>A common wrong answer drops the intercept; C is the actual data value, not the prediction. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P2 | Hard | For the data shown, the line of best fit is y = 2.9x + 2 and the actual value at x = 4 is 14. What is the residual at x = 4?<br>  | 0.4 | Residual = actual − predicted = 14 − 13.6 = **0.4**.<br>Positive, so the point sits **above** the line. |
| P3 | Hard | A scatterplot of hours studied against test score shows points rising steeply and lying very close to a straight line. Which best describes the association?<br>  | Strong positive linear | Rising is positive; points close to the line make it strong; straight makes it linear.<br>**Strong positive linear.** A common wrong answer would need a visible curve. |
| P4 | Hard | In the model y = 2.9x + 2 relating x hours of training to y units produced, what is the best interpretation of 2.9?<br>  | The predicted rise in output for each extra hour of training. | A slope is a **rate of change**: per extra unit of x, y is predicted to rise by 2.9.<br>A common wrong answer describes the intercept. |
| P5 | Hard | A table shows y = 3, 6, 12, 24 as x = 1, 2, 3, 4. Which type of model best fits the data?<br>  | Exponential, because the ratios are constant | Differences are 3, 6, 12 — not constant. Ratios are 2, 2, 2 — constant.<br>A constant ratio means **exponential**. The pairing of reason to conclusion is what is being tested. |
| P6 | Hard | A town's population of 1200 decreases by 8% each year. To the nearest whole number, what is the population after 5 years?<br>  | 791 | A fall of 8% multiplies by 0.92 each year. $1200(0.92)^5 = 790.89... \approx$ **791**. Subtracting 8% of 1200 five times would give 720, which is the common error. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P7 | Very hard | A study finds that towns with more bookshops have higher average exam results. State the best-supported conclusion.<br>  | **There is an association, but a third factor such as household income may explain both.** | The study is **observational**, so it supports association only. A lurking variable such as income can raise both.<br>A causal claim would need random assignment, which no town study can provide. |
| P8 | Very hard | For a set of data the line of best fit is y = 2.9x + 2. At x = 3 the actual value is 10. What is the residual, and what does its sign mean?<br>  | **−0.7; the point is below the line** | Predicted = 2.9(3) + 2 = 10.7. Residual = 10 − 10.7 = **−0.7**.<br>Negative means the actual value is **below** the prediction. A common wrong answer reverses the subtraction. |
| P9 | Very hard | The data used to build the model y = 2.9x + 2 covers x from 1 to 4. A manager uses it to predict the value at x = 30. Which is the best criticism?<br>  | **x = 30 is far outside the data range, so the prediction is unreliable.** | 2.9(30) + 2 = 89 is arithmetically fine, but x = 30 is well beyond the evidence.<br>This is **extrapolation**, and nothing in the data shows the pattern continues. |
| P10 | Very hard | A residual plot for a linear model shows the residuals forming a clear U shape rather than scattering randomly. What does this indicate?<br>  | **A linear model is not appropriate for this data.** | A pattern in the residuals means the model is **missing structure** in the data.<br>A line is the wrong model here — the relationship curves. A good residual plot looks like random scatter about zero. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P11 | Very hard | Two quantities are recorded: y = 100, 90, 81, 72.9 as x = 0, 1, 2, 3. A student says the data is linear because y keeps falling by about the same amount. Why is the student wrong, and what model fits?<br>  | **The differences are −10, −9, −8.1, which are not constant; the ratios are all 0.9, so the model is exponential decay.** | Differences −10, −9, −8.1 shrink; ratios 90/100 = 81/90 = 72.9/81 = 0.9 are constant.<br>A constant ratio below 1 is **exponential decay**, a 10% fall each step. Over a short run a decay curve can look almost straight, which is exactly the trap. |
| P12 | Very hard | Data points (2, 11) and (6, 23) lie on a line of best fit. What does the model predict at x = 10?<br>  | **35** | Slope = (23 − 11)/(6 − 2) = 12/4 = 3, and 11 = 3(2) + b gives b = 5.<br>At x = 10: 3(10) + 5 = **35**. |

**The one habit to build.** A line of best fit is evidence **only inside the range it was built from**, and only ever about **association**. Two of these twelve are answered by refusing to overclaim — and that refusal is the correct answer, not a dodge.

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*

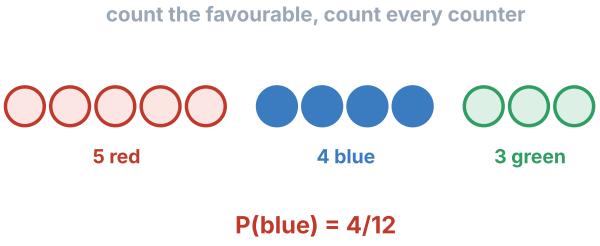
# SAT Math 3.6 — Probability

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Basic probability

**P(event) = favourable ÷ total | every probability lies between 0 and 1**

Count what you want, count everything, divide. The commonest error is **counting the total wrong** — add all the categories, including the one you are not choosing. A probability above 1 always means the denominator is wrong.



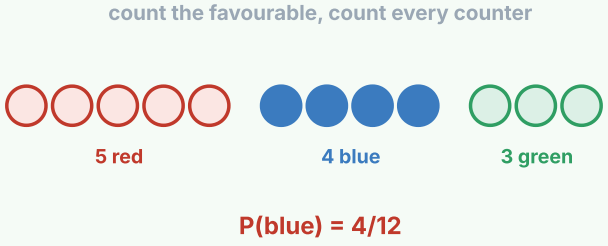

Twelve counters drawn out, so the denominator is counted rather than guessed. Four of them are blue.

**Example 1.** **A bag holds 5 red, 4 blue and 3 green counters. What is P(blue)?**

**Solution.**

Total = 5 + 4 + 3 = 12.

P(blue) = 4/12 = **1/3**.

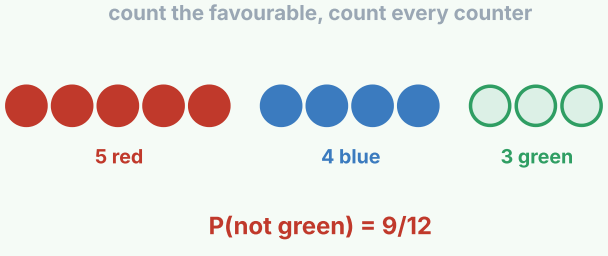



**Example 2.** **For the same bag, what is P(not green)?**

**Solution.**

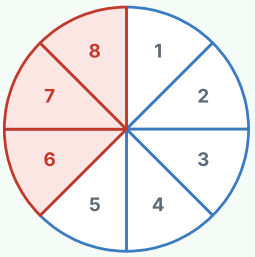
Not green means red or blue: 5 + 4 = 9 of 12.

9/12 = **3/4**. Equivalently 1 − 3/12.



**Example 3.** **A spinner has 8 equal sectors numbered 1 to 8. What is P(a number greater than 5)?**

**Solution.**
The values 6, 7, 8 qualify — three of them.
P = **3/8**. ‘Greater than 5’ excludes 5 itself.



P = 3/8

## 2 Probability from a two-way table

**decide the whole first: grand total, one row, or one column**

Almost every two-way-table question is easy arithmetic and **one hard decision** — which total goes underneath. Read the sentence for the group being described; that group is the denominator.

Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

Everything you need is in the margins. The only question left is which of those margin numbers is the whole.

**Example 1.** **From the table above, what is the probability that a randomly chosen customer bought cheese?**

**Solution.**
The Cheese column totals 60 + 20 = 80, out of 180.
P = 80/180 = **4/9**.

Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

**Example 2.** **What is the probability that a customer bought both fruit and cheese?**

**Solution.**

One cell, 60, out of the grand total 180.

P = 60/180 = **1/3**.

Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

**Example 3.** **What is the probability that a customer bought fruit but not cheese?**

**Solution.**

That cell holds 40, out of 180.

P = 40/180 = **2/9**.

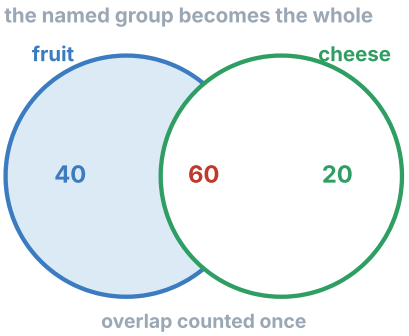
Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

## 3 Conditional probability

**‘given that’ or ‘of those who’ → the named group becomes the whole**

This is the **highest-value idea in the topic** and the one your class loses marks on. The phrase ‘of the customers who bought fruit’ replaces the grand total with the fruit row. Nothing else changes — the numerator is still the overlap.



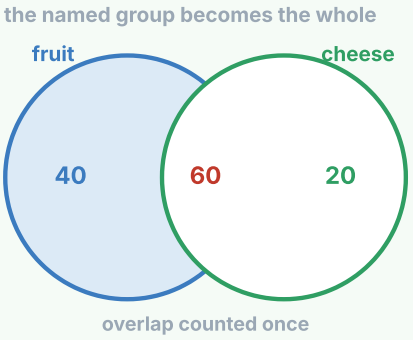

P(cheese | fruit) uses only the shaded left circle as the whole: 60 out of 100, not 60 out of 180.

**Example 1.** **Of the customers who bought fruit, what is the probability one also bought cheese?**

**Solution.**

The whole is now the Fruit row, 100.

P = 60/100 = **3/5**, not 60/180.

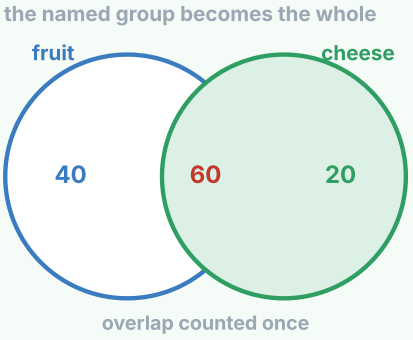



**Example 2.** **Given that a customer bought cheese, what is the probability they bought fruit?**

**Solution.**

Now the whole is the Cheese column, 80.

P = 60/80 = **3/4**. Notice this differs from the previous answer — order matters.



**Example 3.** **Of the customers who did not buy fruit, what fraction bought cheese?**

**Solution.**

The No-fruit row totals 20 + 60 = 80.

P = 20/80 = **1/4**.

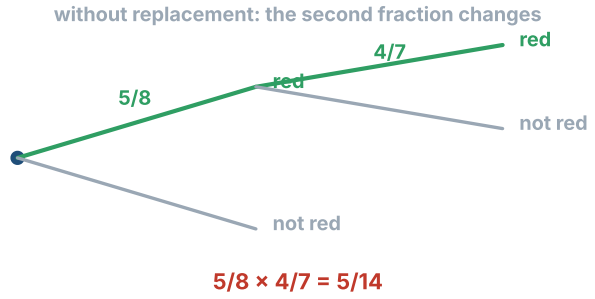
Survey of 180 market customers

| | Cheese | No cheese | Total |
|---|---|---|---|
| Bought fruit | 60 | 40 | 100 |
| No fruit | 20 | 60 | 80 |
| Total | 80 | 100 | 180 |

## 4 Independent events and ‘and’

**independent events: P(A and B) = P(A) × P(B) | without replacement the second probability changes**

**Multiply** along a sequence of events. The question to ask is whether the first event changes the second: drawing **with** replacement leaves the bag unchanged, drawing **without** replacement removes one item from both the numerator and the total.



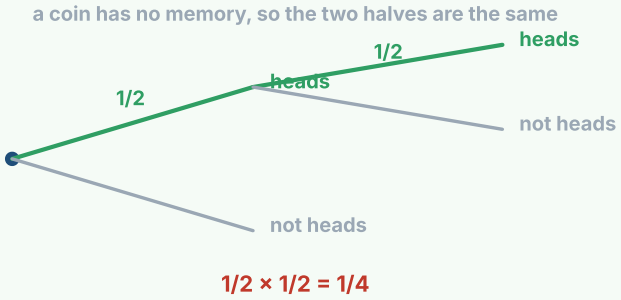

Follow one branch and multiply along it. The second fraction is 4/7, not 5/8, because a red counter has already gone.

**Example 1.** **A fair coin is tossed twice. What is P(two heads)?**

**Solution.**

The tosses are independent.

P = ½ × ½ = **1/4**.

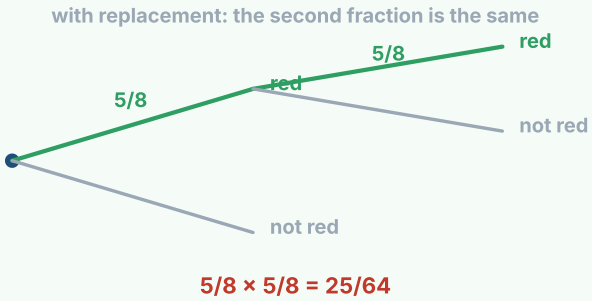


**Example 2.** **A bag has 5 red and 3 blue. Two are drawn with replacement. What is P(both red)?**

**Solution.**

The bag is unchanged between draws.

P = (5/8)(5/8) = **25/64**.

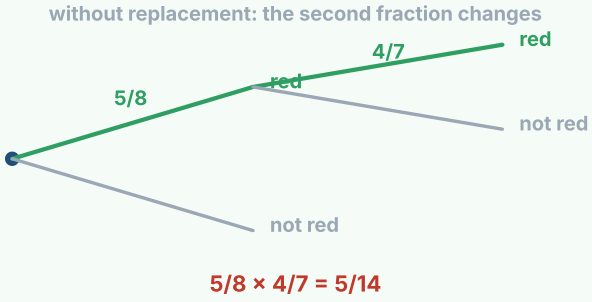



**Example 3.** **The same bag, but drawn without replacement. What is P(both red)?**

**Solution.**

After one red is taken, 4 reds remain out of 7.

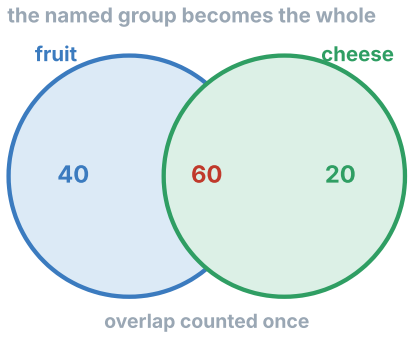
P = (5/8)(4/7) = 20/56 = **5/14**.



## 5 ‘Or’ and the overlap

**P(A or B) = P(A) + P(B) − P(A and B)**

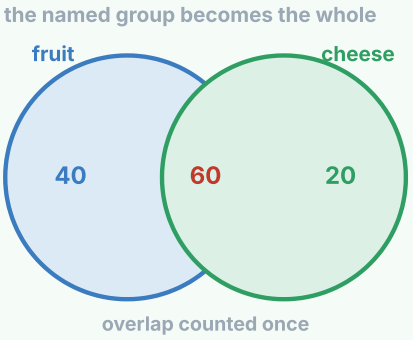
Adding two probabilities **double-counts** anything in both, so subtract the overlap once. If the events cannot happen together, the overlap is zero and the formula collapses to simple addition.




Shading both circles covers 40 + 60 + 20 = 120 people. Adding 100 and 80 would count the middle 60 twice.

**Example 1.** **From the table, what is P(bought fruit or bought cheese)?**

**Solution.**
100/180 + 80/180 − 60/180 = 120/180 = **2/3**. Without subtracting the 60, you would wrongly get 1.



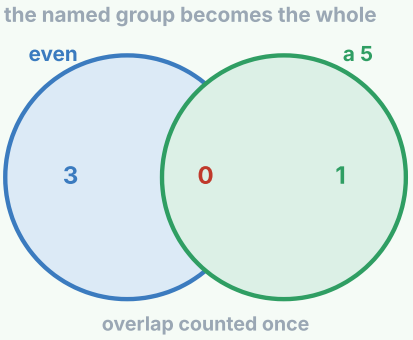

**Example 2.** **A card is drawn from 52. What is P(a heart or a king)?**

**Solution.**
13/52 + 4/52 − 1/52 (the king of hearts) = 16/52 = **4/13**.



**Example 3.** **A die is rolled. What is P(an even number or a 5)?**

**Solution.**
Even gives 3 outcomes, 5 gives 1, and they cannot both happen.
P = 3/6 + 1/6 = 4/6 = **2/3**.




## 6 Expected counts

**expected number = probability × group size**

When a question scales a proportion up to a larger population, **multiply**. This is the same arithmetic as a percentage of a total, and the answer should be rounded to a whole number of people or items when that is what is being counted.




a probability becomes a count the moment you multiply

The fraction is a rate; the group size turns it into people. Nothing more complicated than that.

**Example 1.** **The probability a customer buys cheese is 4/9. In a day with 450 customers, how many are expected to buy cheese?**

**Solution.**
(4/9)(450) = **200 customers**.




**Example 2.** **A machine produces faulty parts 3% of the time. How many faults are expected in 1,200 parts?**

**Solution.**
0.03 × 1200 = **36 parts**.



**Example 3.** **In a survey, 60 of 100 fruit buyers also bought cheese. In a town with 2,500 fruit buyers, how many are expected to buy cheese?**

**Solution.**
The rate is 60/100 = 0.6.
0.6 × 2500 = **1,500 people**.



## Part 1 — With Desmos (10 questions · every one is faster on the graph)

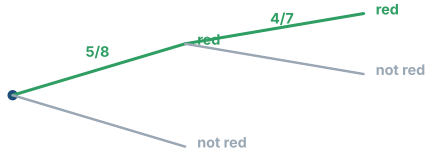
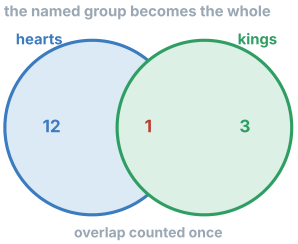
The SAT gives you Desmos on every maths question. Here it is **a fraction calculator**. Type the numerator and denominator you have chosen — the whole difficulty of this topic is choosing the denominator, and Desmos cannot do that.

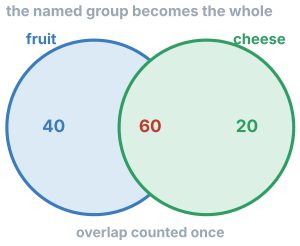
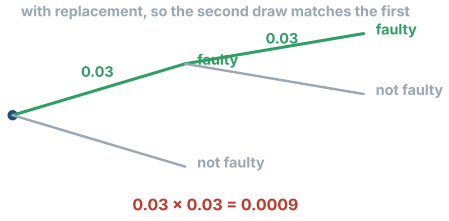
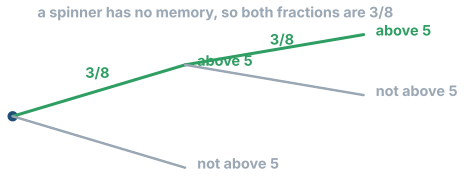
| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A bag holds 5 red, 4 blue and 3 green counters. What is the probability of drawing a blue counter, as a decimal to three places? | `4/12` | Desmos keeps the exact fraction 1/3. | 0.333 |
| D2 | In a survey of 180 customers, 80 bought cheese. What is the probability a randomly chosen customer bought cheese? | `80/180` | Desmos simplifies the fraction for you. | 4/9 |
| D3 | Of 100 fruit buyers, 60 also bought cheese. What is this conditional probability? | `60/100` | The named group, 100, is the denominator. | 0.6 |
| D4 | A fair coin is tossed twice. What is the probability of two heads? | `(1/2)*(1/2)` | Independent events multiply. | 0.25 |
| D5 | A bag has 5 red and 3 blue. Two are drawn without replacement. What is P(both red)? | `(5/8)*(4/7)` | The second fraction changes because one red has gone. | 5/14 |
| D6 | What is the probability that a card drawn from 52 is a heart or a king? | `13/52+4/52-1/52` | Subtract the king of hearts once. | 4/13 |
| D7 | The probability a customer buys cheese is 4/9. How many of 450 customers are expected to? | `(4/9)*450` | Probability times group size. | 200 |
| D8 | A machine is faulty 3% of the time. How many faults are expected in 1,200 parts? | `0.03*1200` | Write the percentage as a decimal. | 36 |
| D9 | From a two-way table, 120 of 180 customers bought fruit or cheese. What is that probability? | `120/180` | Desmos reduces it. | 2/3 |
| D10 | A spinner has 8 equal sectors numbered 1 to 8. What is P(greater than 5)? | `3/8` | Count 6, 7, 8 — not 5 itself. | 0.375 |

## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. One question decides almost all of these: **has the group been narrowed?** If the wording says 'given that', 'of the' or 'among', the denominator is a row or a column, not the whole table.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | The table shows the results of a survey of 180 market customers. If one customer is selected at random, what is the probability that the customer bought cheese?<br>Survey of 180 market customers<br>Cheese / No cheese / Total<br>Bought fruit: 60 / 40 / 100<br>No fruit: 20 / 60 / 80<br>Total: 80 / 100 / 180 | 4/9 | The Cheese column totals 60 + 20 = 80, out of the grand total 180.<br>80/180 = **4/9**. A common wrong answer is the single overlap cell 60/180. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P2 | Hard | Using the same table: of the customers who bought fruit, what is the probability that the customer also bought cheese?<br>Survey of 180 market customers<br>\| | Cheese | No cheese | Total \|<br>\| Bought fruit \| 60 \| 40 \| 100 \|<br>\| No fruit \| 20 \| 60 \| 80 \|<br>\| Total \| 80 \| 100 \| 180 \| | 3/5 | ‘Of the customers who bought fruit’ makes the Fruit row, 100, the whole.<br>60/100 = **3/5**. A common wrong answer divides by 180 instead — the single most common error on this topic. |
| P3 | Hard | A bag contains 5 red, 4 blue and 3 green counters. One counter is drawn at random. What is the probability that it is not green?<br>count the favourable, count every counter<br>  <br><br>P(not green) = 9/12 | 3/4 | Not green means 5 + 4 = 9 of the 12 counters.<br>9/12 = **3/4**. A common wrong answer is P(green) itself. |
| P4 | Hard | A bag contains 5 red and 3 blue counters. Two counters are drawn at random without replacement. What is the probability that both are red?<br>without replacement: the second fraction changes<br>  <br><br>5/8 × 4/7 = 5/14 | 5/14 | First draw 5/8; after one red is removed, 4 reds remain of 7.<br>(5/8)(4/7) = 20/56 = **5/14**. A common wrong answer is the answer if the counter is replaced. |
| P5 | Hard | A card is drawn at random from a standard deck of 52. What is the probability that it is a heart or a king?<br>the named group becomes the whole<br>  <br><br>overlap counted once | 4/13 | 13 hearts + 4 kings, minus the king of hearts counted twice.<br>(13 + 4 − 1)/52 = 16/52 = **4/13**. Forgetting to subtract gives 17/52. |
| P6 | Hard | The probability that a customer at a market buys cheese is 4/9. On a day when 450 customers visit, how many would be expected to buy cheese?<br>  <br> | 200 | Expected count = probability × group size.<br>(4/9)(450) = **200 customers**. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P7 | Very hard | Using the survey table, what is the probability that a randomly chosen customer bought fruit or cheese?<br>  | 2/3 | 100/180 + 80/180 would double-count the 60 who did both.<br>(100 + 80 − 60)/180 = 120/180 = **2/3**. A common wrong answer is what you get by adding 100 and 80 and not subtracting the overlap. |
| P8 | Very hard | In the same survey, let A be 'bought cheese given bought fruit' and B be 'bought fruit given bought cheese'. State what is true here.<br>same numerator, different whole — so A ≠ B<br>A = P(cheese \| fruit) 60 ÷ 100 = 0.6<br>B = P(fruit \| cheese) 60 ÷ 80 = 0.75 | A < B, because 100 is larger than 80. | A = 60/100 = 0.6 and B = 60/80 = 0.75, so **A < B**.<br>The numerator is the same 60 both times; only the denominator changes, and the larger denominator gives the smaller probability. A common wrong answer is the classic error of treating conditional probability as symmetric. |
| P9 | Very hard | A quality check finds that 3% of parts are faulty. In a batch of 1,200 parts, two are drawn at random with replacement. What is the probability that both are faulty?<br>  <br>0.03 × 0.03 = 0.0009 | 0.0009 | With replacement the draws are independent.<br>(0.03)(0.03) = **0.0009**. A common wrong answer adds the two probabilities instead of multiplying; the batch size of 1,200 is not needed at all. |
| P10 | Very hard | A spinner has 8 equal sectors numbered 1 to 8. It is spun twice. What is the probability that both spins give a number greater than 5?<br>  <br>3/8 × 3/8 = 9/64 | 9/64 | Each spin: 6, 7, 8 qualify, so 3/8. The spins are independent.<br>(3/8)(3/8) = **9/64**. A common wrong answer is one spin only. |
| P11 | Very hard | In a survey of 180 customers, 100 bought fruit and 60 of those also bought cheese; 20 of the remaining 80 bought cheese. A manager claims that buying fruit and buying cheese are independent. Which calculation best tests the claim?<br>independence is a comparison, not a formula<br>P(cheese \| fruit) 60 ÷ 100 = 0.6<br>P(cheese) overall 80 ÷ 180 ≈ 0.44<br>verdict different, so not independent | Compare 60/100 with 80/180. | Independence means P(cheese \| fruit) equals P(cheese).<br>P(cheese \| fruit) = 60/100 = 0.6 and P(cheese) = 80/180 ≈ 0.44. They differ, so the events are **not independent** and the claim fails. A common wrong answer compares the two conditional probabilities, which tests something else. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P12 | Very hard | Of 100 fruit buyers surveyed, 60 also bought cheese. A town has 2,500 fruit buyers. Assuming the same rate, how many are expected to buy cheese?<br>  | 1500 | The rate is 60/100 = 0.6.<br>0.6 × 2500 = **1,500 people**. |

**The one habit to build.** Your class scored 20% on conditional probability, and the cause is always the same: **the right numerator over the wrong denominator.** Before dividing, say which group you are now inside.

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*

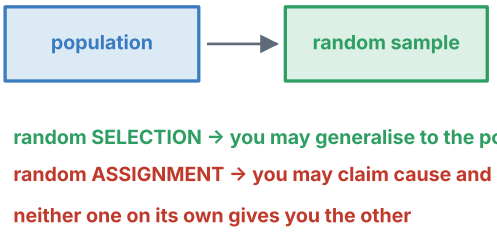
# SAT Math 3.7 — Inference & Margin of Error

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Population, sample and parameter

**the population is everyone you want to describe | the sample is who you actually asked**

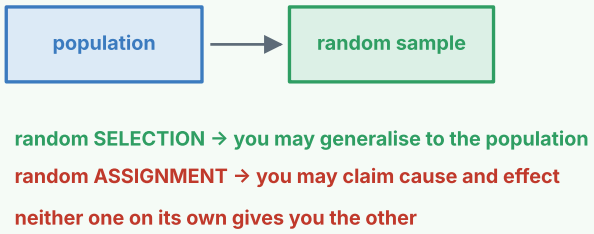
A **statistic** comes from the sample; a **parameter** describes the population. The whole point of sampling is to estimate the parameter you cannot measure. The sample must be drawn from the population you want to describe — surveying one school tells you nothing about a country.




One arrow, and everything downstream depends on it. What the sample licenses you to say is fixed by how it was drawn.

**Example 1.** **A school of 1,200 students is surveyed by asking 150 of them. Name the population and the sample.**

**Solution.**
**Population** = the 1,200 students; **sample** = the 150 asked.




**Example 2.** **In that survey 60% of those asked walk to school. Is 60% a statistic or a parameter?**

**Solution.**
It comes from the sample, so it is a **statistic**.
It estimates the unknown **parameter**, the true percentage among all 1,200.

which of the two did you actually measure?

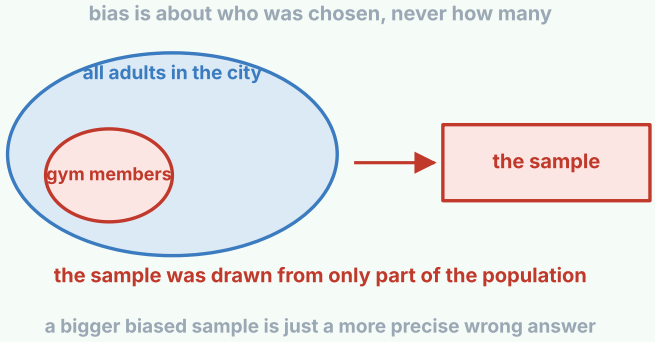
| | |
|---|---|
| statistic | measured on the sample — 60% |
| parameter | describes all 1,200 — unknown |

**Example 3. A researcher wants to describe all adults in a city but surveys only gym members. What is wrong?**

**Solution.**

The sample is **not drawn from the whole population**, so it is biased.

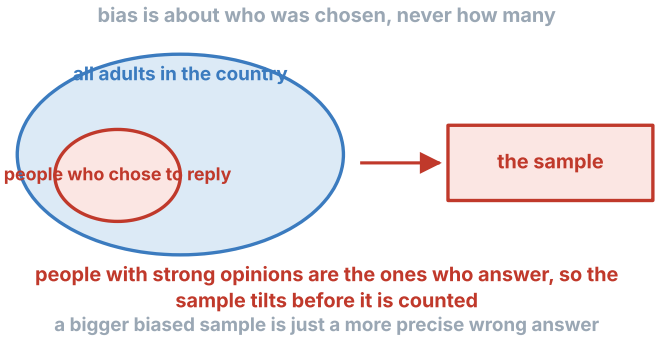
Gym members differ systematically from adults in general, and no sample size fixes that.




## 2 Random selection and bias

**only a random sample supports generalising to the population**

Bias is about **how** people were chosen, not **how many**. A voluntary online poll, a convenience sample at one location, or a survey of people who chose to reply are all biased no matter how large. Increasing a biased sample makes the wrong answer more precise, not more correct.



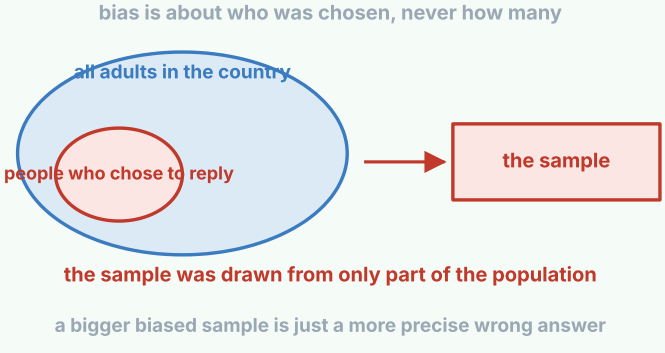

The red blob is who actually answered. It sits inside the population but is not representative of it, and 5,000 replies do not change its position.

**Example 1. A survey is posted online and 5,000 people reply voluntarily. Is it reliable for the population?**

**Solution.**

**No.** Voluntary response is biased — people with strong opinions answer.

5,000 replies do not repair that.



**Example 2.** **Which is better for estimating a school's average: 60 randomly chosen students, or 300 volunteers?**

**Solution.**

**The 60 random students.** Random selection removes systematic bias; volunteering does not.

size cannot buy what randomness gives

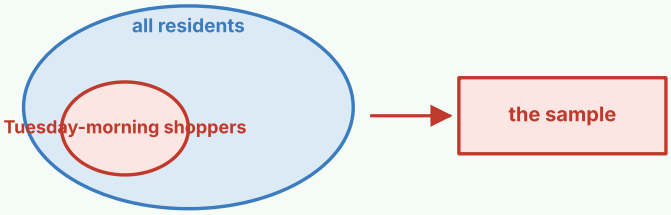
| | |
|---|---|
| 60 chosen at random | no systematic tilt — usable |
| 300 volunteers | tilted, however many there are |

**Example 3.** **A council surveys shoppers at one shopping centre on a Tuesday morning. Name the problem.**

**Solution.**

A **convenience sample**. Tuesday-morning shoppers are not representative of all residents, so the results cannot be generalised.

bias is about who was chosen, never how many



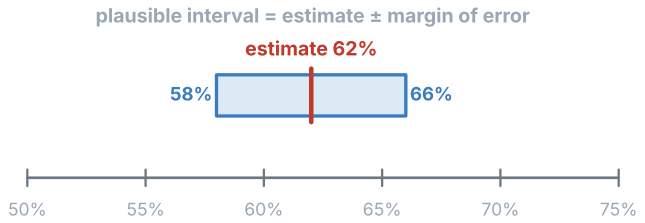

whoever happened to be there is not whoever lives in the town

a bigger biased sample is just a more precise wrong answer

## 3 Margin of error

**plausible interval = estimate ± margin of error**

The margin of error turns a single estimate into a **range of believable values** for the population. The SAT never asks you to compute one from scratch; it asks you to **build** the interval, **read** it, or say **what changes** it.



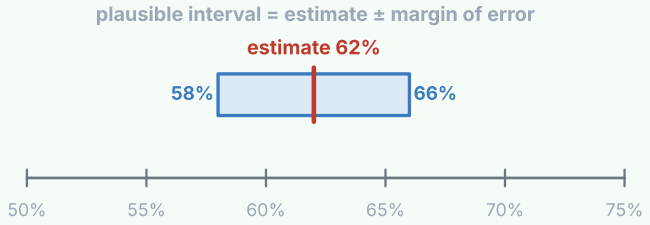

The estimate sits in the middle and the margin runs both ways. Every value inside the bar is a believable population value.

**Example 1.** **A poll estimates 62% with a margin of error of 4 percentage points. Give the plausible interval.**

**Solution.**

62 − 4 = 58 and 62 + 4 = 66.

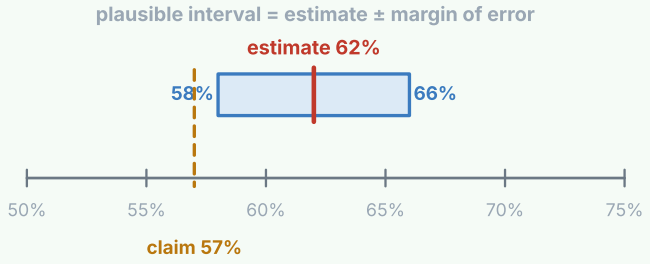
The interval is **58% to 66%**.




**Example 2.** **Is a true value of 57% consistent with that poll?**

**Solution.**

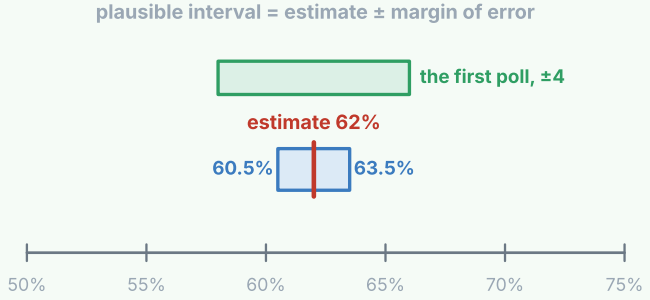
**No.** 57% lies outside 58%–66%, so it is not among the plausible values.




**Example 3.** **A second poll reports 62% with a margin of error of 1.5 points. What is its interval?**

**Solution.**

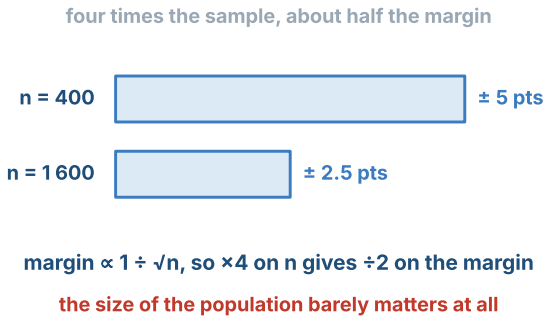
**60.5% to 63.5%** — a narrower, more precise estimate of the same quantity.



## 4 What makes the margin of error smaller

**a larger sample narrows it | a higher confidence level widens it**

**Sample size** is the lever the SAT tests. More data gives a tighter interval, but with diminishing returns — the margin shrinks with the **square root** of the sample size, so quadrupling the sample roughly halves it. Population size is almost irrelevant.



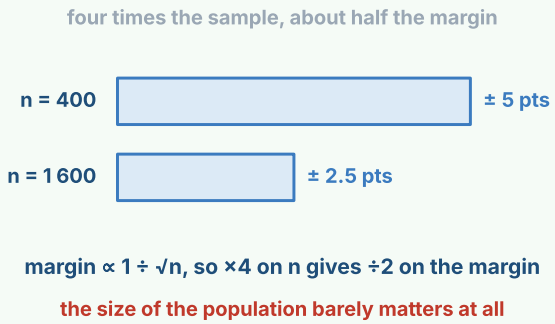

Four times as many people, about half the margin. Sixteen times as many would be needed to quarter it.

**Example 1.** **A survey of 400 gives a margin of error of 5 points. What would 1,600 give, roughly?**

**Solution.**

Four times the sample roughly halves the margin.

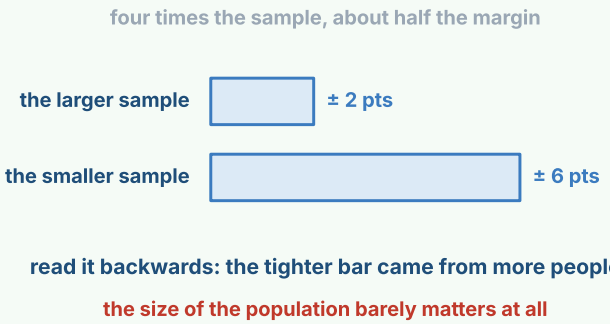
**About 2.5 points.**




**Example 2.** **Two polls of the same population report margins of 2 and 6 points. Which used a larger sample?**

**Solution.**

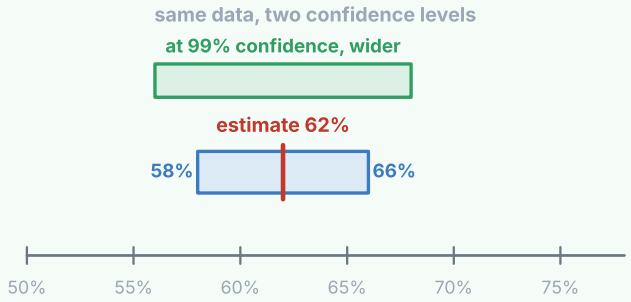
**The one with the 2-point margin** — smaller margin, larger sample.



**Example 3.** **Does raising the confidence level from 95% to 99% make the interval wider or narrower?**

**Solution.**

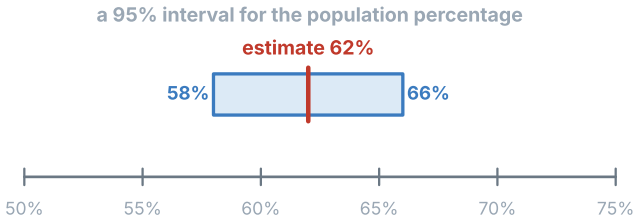
**Wider.** To be more confident of capturing the true value, you must allow a bigger range.




## 5 Reading a confidence statement

**the interval is about the population value, not about individuals**

The **language** matters more than the arithmetic. A 95% confidence interval of 58% to 66% says the plausible values **for the whole population** lie in that range. It does **not** say 95% of people, and it does not say any individual has a 95% chance of anything.

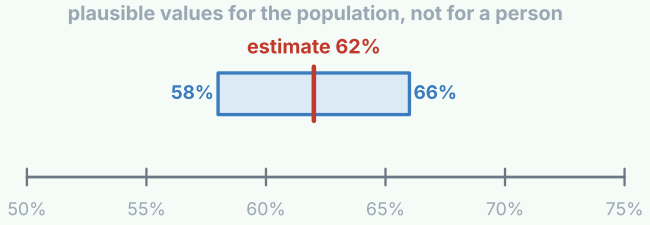



The 95% is a property of the method, not a number on this axis. What the axis shows is the share who walk to school.

**Example 1.** **A poll gives a 95% confidence interval of 58% to 66% for the share who walk to school. State what it means.**

**Solution.**

It is plausible that **the true percentage for the whole population** lies between 58% and 66%.



**Example 2.** **Does it mean 95% of students walk to school?**

**Solution.**

**No.** 95% is the confidence level, not a percentage of students.

The estimate of walkers is **62%**.

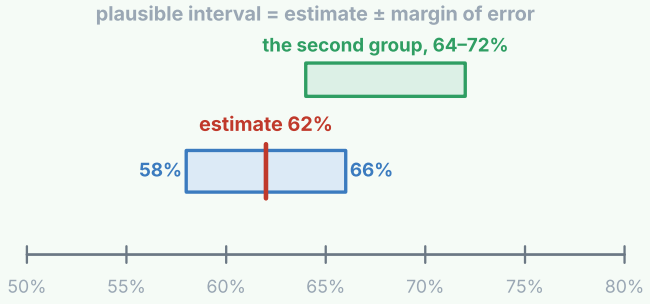
three numbers, three different jobs

| | |
|---|---|
| 95% | the confidence level — a property of the method |
| 62% | the estimated share who walk |
| 58% to 66% | the plausible values for that share |

**Example 3.** **Two groups' intervals are 58–66% and 64–72%. Can you say one group is higher?**

**Solution.**

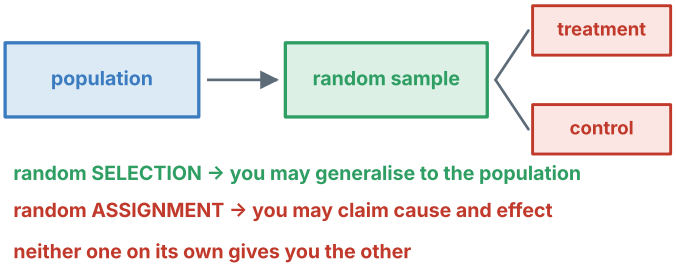
**Not confidently** — the intervals overlap between 64% and 66%, so the same true value is plausible for both.



## 6 Which conclusions a study supports

**random selection → generalise | random assignment → cause**

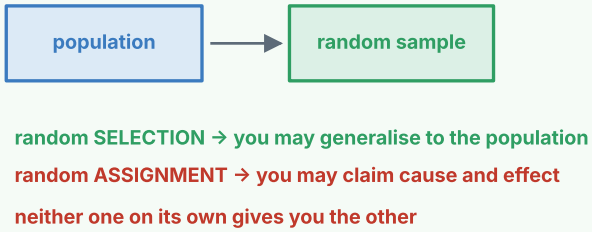
Two different randomisations license **two different conclusions**, and the SAT keeps them apart. Selecting participants at random lets you extend findings to the population. Assigning them to groups at random lets you claim the treatment caused the difference. **You need both for both.**




Two separate random steps. Each buys exactly one conclusion, and neither lends its conclusion to the other.

**Example 1.** **200 students are chosen at random from a school and surveyed. What can you conclude?**

**Solution.**
You may **generalise to that school**, but not to other schools and not about cause.




**Example 2.** **Volunteers are randomly assigned to two study methods and one scores higher. What follows?**

**Solution.**
You may claim the method **caused** the difference for people like these volunteers, but you **cannot generalise** beyond them.

random assignment without random selection

| | |
|---|---|
| how they joined | they volunteered — not random |
| how they were grouped | at random |
| so you may say | cause, for these volunteers only |

**Example 3.** **A study randomly selects participants from a city and randomly assigns them to treatments. What is supported?**

**Solution.**
**Both** a causal claim **and** generalisation to that city — the strongest design the SAT will show you.




random SELECTION → you may generalise to the population

random ASSIGNMENT → you may claim cause and effect

neither one on its own gives you the other

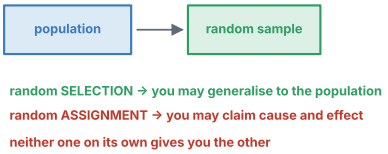
## Part 1 — With Desmos (10 questions · calculator work only; the reasoning is in the wording)

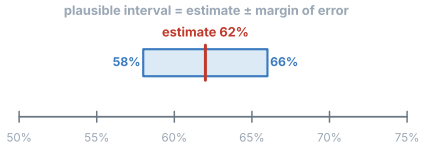
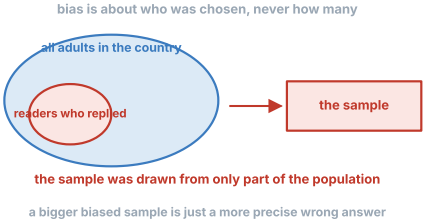
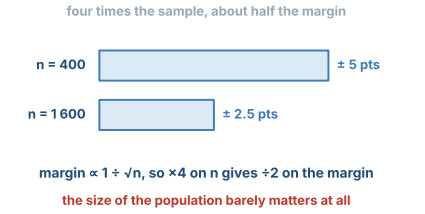
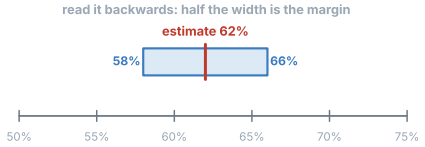
The SAT gives you Desmos on every maths question. Here it does almost nothing: **add and subtract the margin of error**, and scale a rate to a population. Every mark in this topic is in the wording, not the arithmetic.

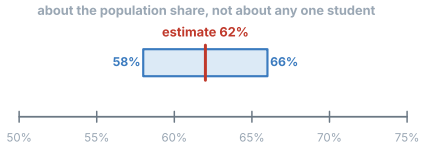
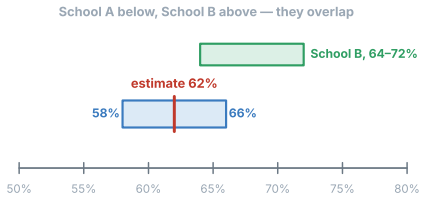
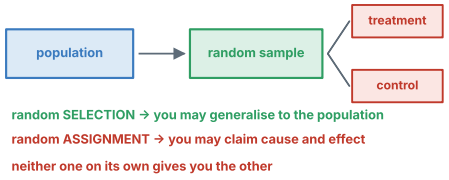
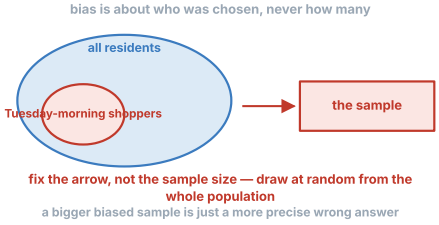
| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A poll estimates 62% with a margin of error of 4 percentage points. What is the lower end of the plausible interval? | `62-4` | Subtract for the lower end. | 58 |
| D2 | For the same poll, what is the upper end? | `62+4` | Add for the upper end. | 66 |
| D3 | A sample of 150 out of 1,200 students is surveyed. What fraction of the population is sampled? | `150/1200` | A sampled fraction, not a probability. | 0.125 |
| D4 | In a sample of 150, 90 students walk to school. What percentage is that? | `90/150*100` | Convert to a percentage at the end. | 60 |
| D5 | A survey of 400 has a margin of error of 5 points. Four times the sample roughly halves it. What is the new margin? | `5/2` | Quadrupling the sample halves the margin. | 2.5 |
| D6 | An interval runs from 60.5% to 63.5%. What is the margin of error? | `(63.5-60.5)/2` | Half the width of the interval. | 1.5 |
| D7 | An interval runs from 58% to 66%. What is the point estimate at its centre? | `(58+66)/2` | The estimate sits in the middle. | 62 |
| D8 | In a town of 24,000 adults, a poll estimates 38% support. How many adults is that? | `0.38*24000` | Scale the proportion up to the population. | 9120 |
| D9 | A sample proportion is 0.44 in a survey of 180 people. How many people is that? | `0.44*180` | Desmos keeps the decimal; round to a whole person if the question asks. | 79.2 |
| D10 | Two polls report margins of 2.0 and 6.0 points. How many times wider is the second interval? | `6/2` | Compare the margins directly. | 3 |

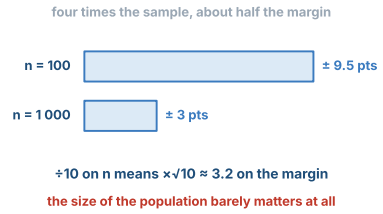
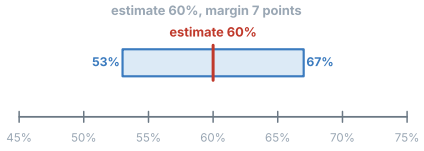
## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Two questions answer nearly all of these: **who was sampled**, and **is the claimed value inside the interval**. Neither needs a calculation.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | A researcher surveys 150 students selected at random from the 1,200 students at a school and finds that 60% walk to school. Which is the parameter of interest?<br>  <br>random SELECTION → you may generalise to the population<br>random ASSIGNMENT → you may claim cause and effect<br>neither one on its own gives you the other | **The percentage of all 1,200 students who walk to school.** | A **parameter** describes the population. The 60% is a statistic that estimates it.<br>The groups in B and D are the sample and the population themselves, not percentages. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P2 | Hard | A poll of registered voters estimates that 62% support a proposal, with a margin of error of 4 percentage points. Which interval gives the plausible values for the whole population?<br>  | **58% to 66%** | Estimate ± margin of error: 62 − 4 to 62 + 4.<br>**58% to 66%.** Two common wrong answers use the margin on one side only. |
| P3 | Hard | A website posts a survey and 5,000 readers choose to respond. Which is the strongest objection to generalising the results to all adults in the country?<br>  | **Respondents chose themselves, so the sample is biased.** | This is **voluntary response bias**: people with strong views reply.<br>5,000 is a large sample, so A is wrong — and a large biased sample is still biased. |
| P4 | Hard | A survey of 400 people produces a margin of error of 5 percentage points. Approximately what margin of error would a survey of 1,600 similar people produce?<br>  | **2.5** | The margin shrinks with the square root of the sample size, so four times the sample roughly halves it.<br>**About 2.5 points.** A common wrong answer divides by 4 instead of by 2. |
| P5 | Hard | A 95% confidence interval for the percentage of residents who recycle runs from 58% to 66%. What is the margin of error, in percentage points?<br>  | **4** | The interval is the estimate ± the margin, so the margin is half its width.<br>(66 − 58)/2 = **4 percentage points**. |
| P6 | Hard | In a town of 24,000 adults, a random sample estimates that 38% support a new library. Approximately how many adults in the town does this estimate represent?<br>  | **9120** | 0.38 × 24,000 = **9,120 adults**. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P7 | Very hard | A study reports a 95% confidence interval of 58% to 66% for the share of students who walk to school. State the best-supported conclusion.<br>  | **It is plausible that the true percentage for all students is between 58% and 66%.** | The interval describes plausible values for the **population percentage**.<br>A common wrong answer confuses the confidence level with the quantity measured; B treats an estimate as exact. |
| P8 | Very hard | Two schools are surveyed. School A's interval for the proportion who walk is 58% to 66%; School B's is 64% to 72%. What can be concluded?<br>  | **The intervals overlap, so the data does not establish which school is higher.** | The intervals share the range 64%–66%, so a single true value could sit in both.<br>No confident difference can be claimed. **Overlap is the thing to look for.** |
| P9 | Very hard | Researchers randomly select 200 adults from a city and randomly assign them to one of two exercise programmes. Programme A's group shows greater improvement. State the best-supported conclusion.<br>  | **Programme A causes greater improvement for adults in that city.** | Random **assignment** supports the causal claim; random **selection** from the city supports generalising to that city — but no further.<br>A common wrong answer over-reaches to a population never sampled. |
| P10 | Very hard | A council wants to estimate the average commute time of all its residents. It surveys shoppers at one shopping centre on a Tuesday morning. Which change would most improve the study?<br>  | **Select residents at random from the council's records.** | The flaw is **how** people were chosen, not **how many**.<br>Only random selection from the whole population removes it; A makes a biased estimate more precise, not more accurate. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P11 | Very hard | A poll of 1,000 people reports 45% support with a margin of error of 3 points. The pollster then reports 45% with a margin of error of 3 points for a subgroup of only 100 of those people. What is wrong?<br>  | **The subgroup is ten times smaller, so its margin of error must be larger — roughly 9.5 points.** | Margin of error depends on **sample size**. A tenth of the sample widens the margin by a factor of about √10 ≈ 3.2, so roughly 3 × 3.2 ≈ **9.5 points**.<br>Reusing the full-sample margin for a subgroup overstates the precision badly. |
| P12 | Very hard | A survey of a random sample of 150 of a school's 1,200 students finds 90 walk to school, with a margin of error of 7 percentage points. What is the lower end of the plausible interval for the whole school, as a percentage?<br>  | **53** | The sample estimate is 90/150 = 60%.<br>60 − 7 = **53%**. The population size of 1,200 plays no part in the calculation. |

**The one habit to build.** A margin of error covers **sampling variation and nothing else**. It does not cover a biased sample, a leading question or people answering dishonestly — and no sample size will fix those.

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*

# SAT Math 3.8 — Evaluating Statistical Claims

Explanations with worked examples · diagrams for every rule · Desmos questions | **Teacher copy — with solutions** · Grade 11 · Ellipse International School

## 1 Observational study or experiment?

**a study that only watches cannot show cause | an experiment imposes a treatment**

Ask one question: **did the researchers decide who got what?** If they merely recorded what people already did, it is **observational** and the strongest claim available is association. If they assigned the treatment, it is an **experiment** and cause is on the table.




**random SELECTION → you may generalise to the population**

**random ASSIGNMENT → you may claim cause and effect**

**neither one on its own gives you the other**

The second random step is the one that creates an experiment. Without it there is no treatment, and so no cause to claim.

**Example 1.** **Researchers record the diets of 500 adults and their blood pressure. Which type of study is this?**

**Solution.**
**Observational.** Nobody was told what to eat, so no causal claim is supported.

observational: the researchers only watched

| | | |
|---|---|---|
| was a treatment imposed? | no — people ate as they already did | × no |
| so the claim available is | association only | ✓ yes |

**Example 2.** **Researchers randomly give 250 adults diet A and 250 diet B, then measure blood pressure. Which type?**

**Solution.**
**An experiment** — the treatment was imposed, and at random.



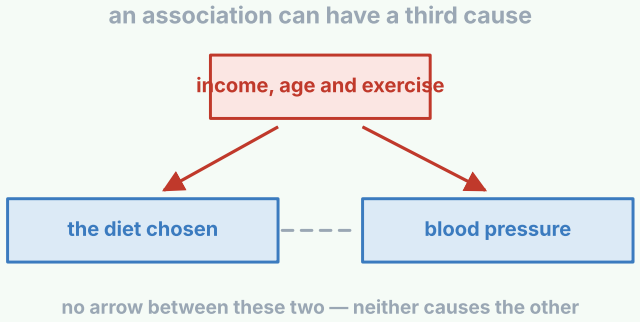

**random SELECTION → you may generalise to the population**

**random ASSIGNMENT → you may claim cause and effect**

**neither one on its own gives you the other**

**Example 3.** **Why can the first study not show that diet causes high blood pressure?**

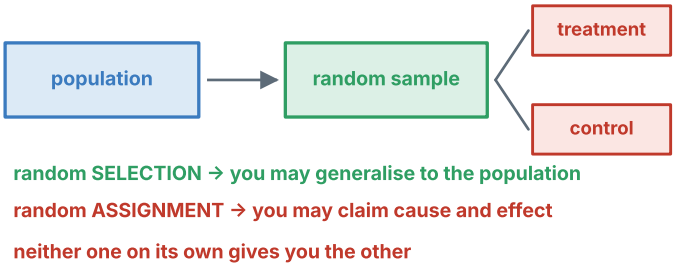
**Solution.**
People who choose a diet differ in other ways too — age, income, exercise.
Any of these **confounding variables** could explain the difference.




## 2 Random selection versus random assignment

**random selection → generalise to the population | random assignment → claim cause**

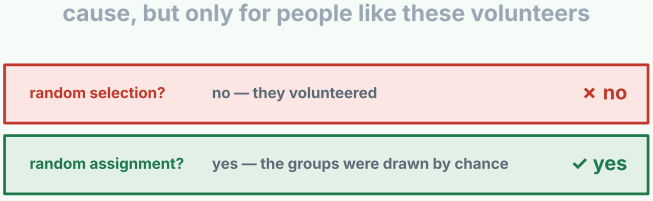
These two are **constantly swapped** on the SAT. **Selection** is how people entered the study; **assignment** is how they were split once inside. Each buys exactly one conclusion, and you need both to get both.




Two arrows, two licences. Cover either one and the conclusion it bought disappears with it.

**Example 1.** **Volunteers are randomly assigned to two tutoring methods. What can and cannot be concluded?**

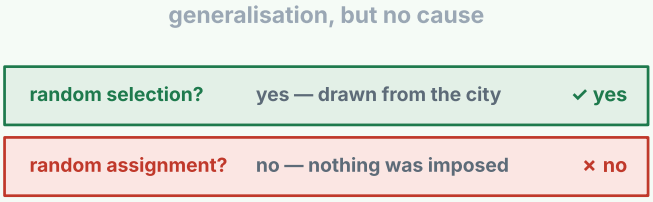
**Solution.**
**Can:** the method caused the difference, for people like these volunteers.
**Cannot:** generalise to all students — the volunteers were not randomly selected.



**Example 2.** **A random sample of a city's residents is surveyed about commuting. What is supported?**

**Solution.**

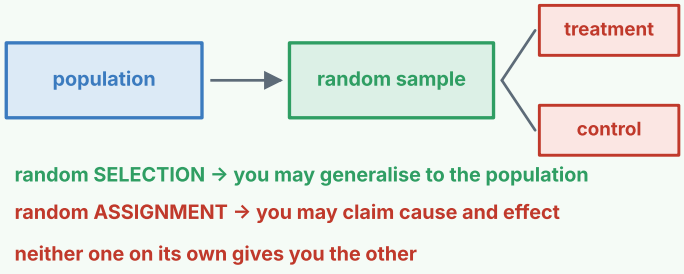
**Generalisation to that city's residents.** No treatment was imposed, so no causal claim.




**Example 3.** **Which design supports both conclusions?**

**Solution.**

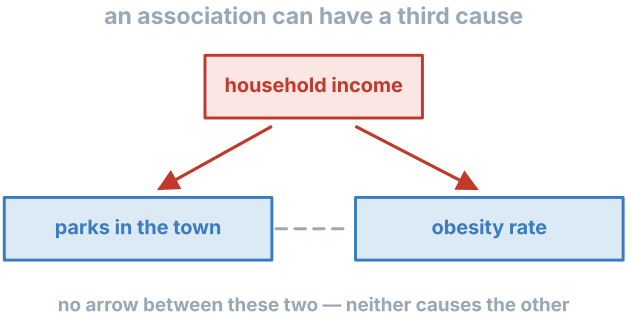
**Random selection from the population and random assignment to groups** — a randomised experiment on a random sample.



## 3 Confounding variables

**a third variable linked to both the cause and the effect**

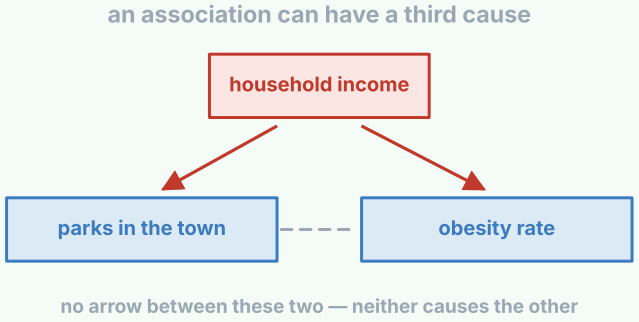
When a question asks why an observational result may mislead, the answer is usually a **named third variable**. A good answer identifies something plausibly connected to **both** quantities — not just any difference between the groups.




A confounder needs an arrow to each of the two quantities. One arrow is not enough, which is how the wrong options are built.

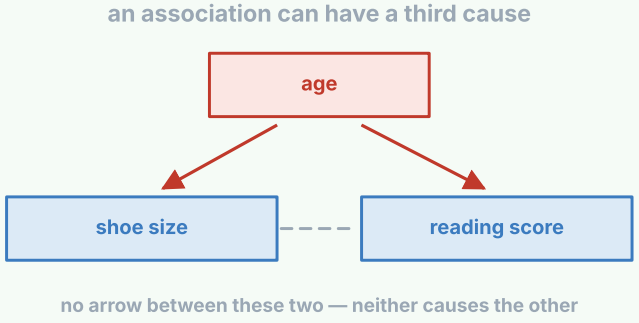
**Example 1.** **Towns with more parks have lower obesity rates. Suggest a confounding variable.**

**Solution.**
**Household income.** Wealthier areas tend to have both more parks and lower obesity, so the parks may not be the cause.




**Example 2.** **Children with larger shoe sizes read better. What explains it?**

**Solution.**
**Age.** Older children have bigger feet and read better; shoe size causes nothing.



**Example 3.** **Why does random assignment remove confounding?**

**Solution.**
It spreads the other variables **evenly across the groups on average**, so the groups differ only in the treatment.




random SELECTION → you may generalise to the population

random ASSIGNMENT → you may claim cause and effect

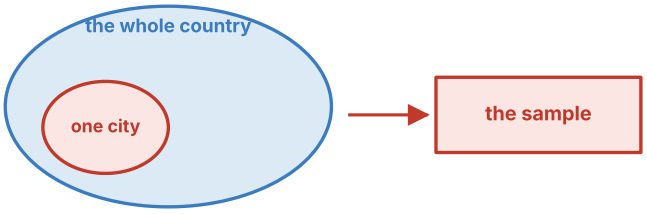
neither one on its own gives you the other

## 4 How the sample limits the conclusion

**you may generalise only to the population the sample was drawn from**

Match the conclusion to the **sampling frame**. A random sample of one school supports claims about **that school** and nothing wider. The commonest wrong answer on these questions is a true-sounding statement about a population that was never sampled.

bias is about who was chosen, never how many




the conclusion may reach the circle the sample came from, and no further

a bigger biased sample is just a more precise wrong answer

The sample defines its own ceiling. Everything outside the red blob is a population nobody asked.

**Example 1.** **200 students are chosen at random from one school. To whom do the results apply?**

**Solution.**
**To that school's students only** — not to the district, and not to students generally.



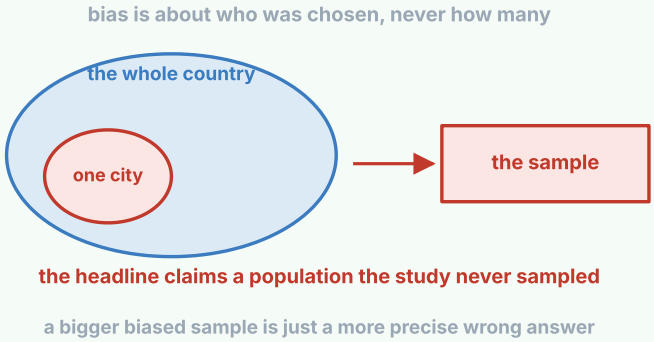

random SELECTION → you may generalise to the population

random ASSIGNMENT → you may claim cause and effect

neither one on its own gives you the other

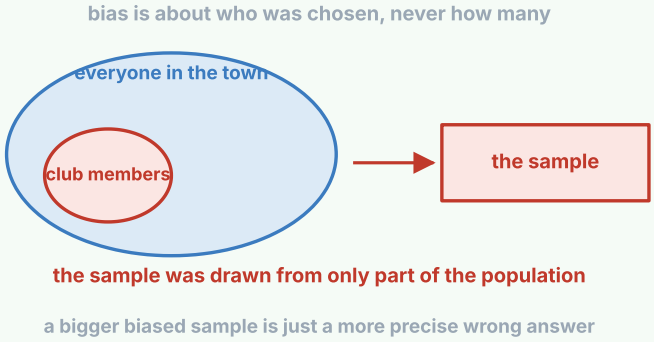
**Example 2. A study samples randomly from all adults in one city. A newspaper reports the result as a national finding. What is wrong?**

**Solution.**
The sampling frame was **one city**, so the conclusion cannot be stretched to the country.




**Example 3. A sample is drawn at random from members of a sports club. Can the result describe the town?**

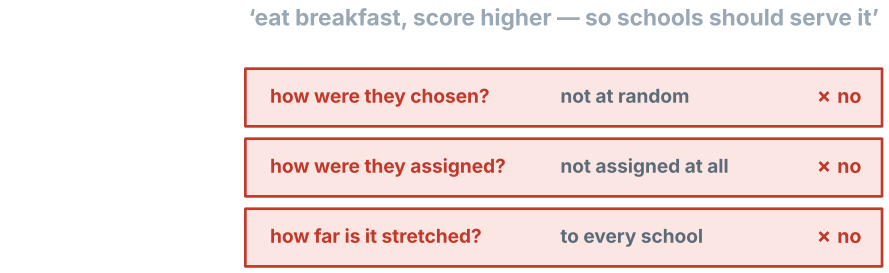
**Solution.**
**No.** The population is the club's members; club members differ systematically from the town as a whole.



## 5 Judging a reported claim

**check three things: how chosen, how assigned, how far stretched**

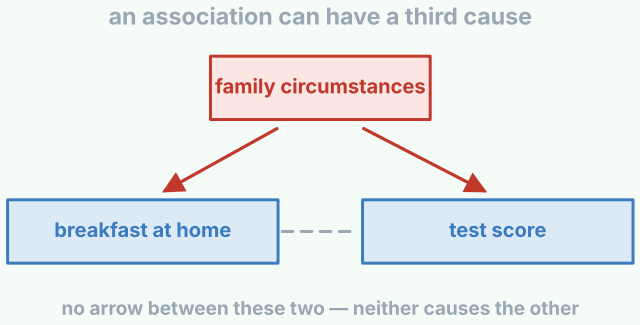
Most of these questions give you a headline and four conclusions, and **the right one is the cautious one**. Scan for the overreach: a causal verb on an observational study, or a population wider than the one sampled.




Three boxes, and this headline fails all three. The correct answer to such a question is always the one that claims least.

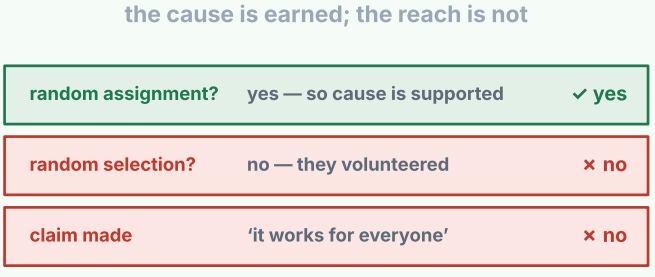
**Example 1.** **Headline: 'Students who eat breakfast score higher — so schools should serve breakfast.' Evaluate it.**

**Solution.**
The data is **observational**, so it shows association only.
Families who provide breakfast may differ in other ways. **The recommendation outruns the evidence.**




**Example 2.** **A randomised trial on 300 volunteers finds a drug works. A report says it works for everyone. What is the flaw?**

**Solution.**
Random **assignment** supports cause within the trial, but the volunteers were **not randomly selected**, so the result cannot be generalised to everyone.



**Example 3. What one change would let the breakfast study support its recommendation?**

**Solution.**

**Randomly assign** students to receive breakfast or not, then compare — turning the observational study into an experiment.




random SELECTION → you may generalise to the population

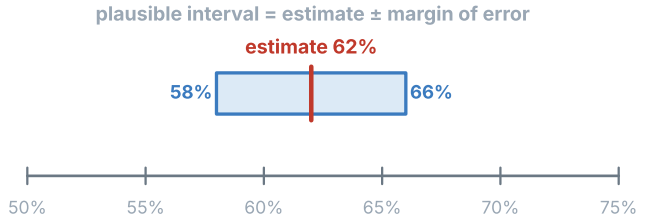
random ASSIGNMENT → you may claim cause and effect

neither one on its own gives you the other

## 6 Precision, bias and sample size

**a bigger sample narrows the margin of error | only better selection removes bias**

Sample size and bias solve **different problems** and the SAT tests whether you know which. A larger sample makes the estimate more **precise**; it does nothing about a sample drawn from the **wrong people**.

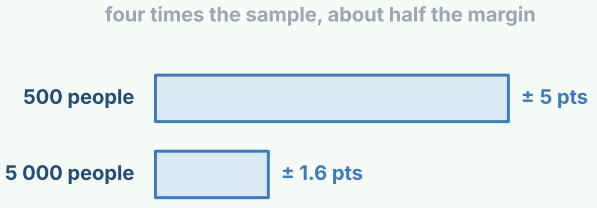



Sample size moves the ends of this bar inwards. Nothing about sample size moves the bar to where it should have been.

**Example 1. A biased survey of 500 is repeated with 5,000 people the same way. What improves?**

**Solution.**

**Precision only.** The margin of error narrows, but the estimate is still centred on the wrong value.



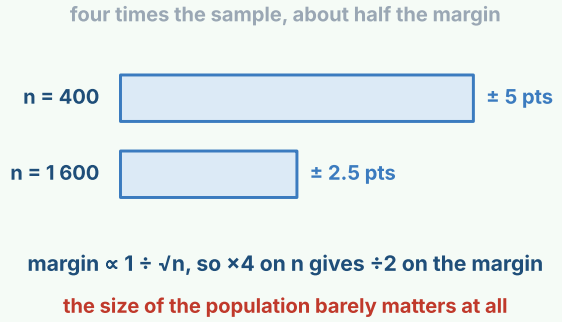

narrower, and still centred in the wrong place

the size of the population barely matters at all

**Example 2.** **A survey of 400 gives a margin of 5 points. What does 1,600 give, roughly?**

**Solution.**

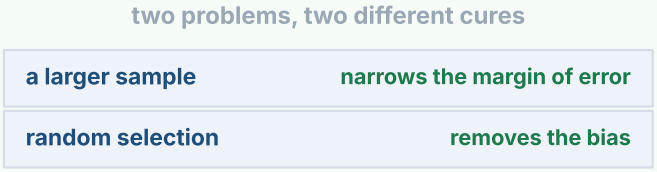
**About 2.5 points** — four times the sample roughly halves the margin.




**Example 3.** **Which fixes bias: a larger sample, or random selection?**

**Solution.**

**Random selection.** Size addresses precision; selection addresses bias, and no amount of size repairs bad selection.



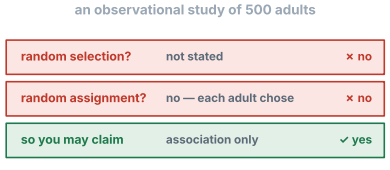
## Part 1 — With Desmos (10 questions · calculator work only; the judgement is in the wording)

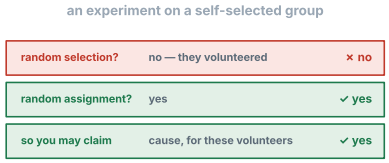
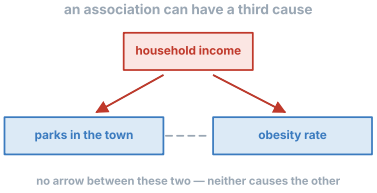
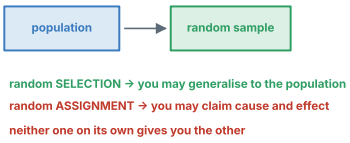
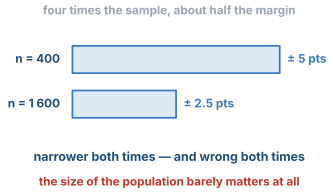
The SAT gives you Desmos on every maths question. Here it is **a calculator for the percentages inside the question**. The decision — what the study is allowed to claim — is entirely yours.

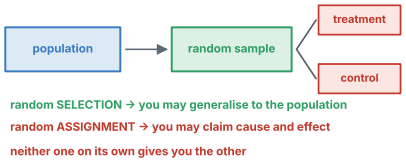
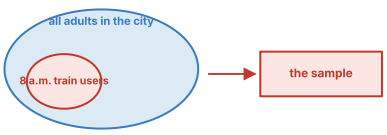
| # | Question | Type into Desmos | Then | Answer |
|---|---|---|---|---|
| D1 | A survey of 400 people has a margin of error of 5 percentage points. Four times the sample roughly halves it. What is the new margin? | `5/2` | Quadrupling the sample halves the margin. | 2.5 |
| D2 | A random sample of 250 of a city's 40,000 adults is surveyed. What fraction of the population is sampled? | `250/40000` | A sampled fraction. | 0.00625 |
| D3 | In an experiment, 250 adults receive diet A and 250 receive diet B. How many take part in all? | `250+250` | Group sizes add. | 500 |
| D4 | A trial reports 38% improvement among 300 volunteers. How many people improved? | `0.38*300` | Proportion times group size. | 114 |
| D5 | An estimate of 62% has a margin of error of 4 points. What is the width of the plausible interval, in percentage points? | `2*4` | The interval spans the margin on each side. | 8 |
| D6 | Of 500 adults studied, 180 reported a high-salt diet. What percentage is that? | `180/500*100` | Convert at the end. | 36 |
| D7 | A study of 1,200 students samples 150. By what factor would the sample need to grow to halve the margin of error? | `4` | Margin shrinks with the square root of the sample size. | 4 |
| D8 | Two groups of 250 report mean improvements of 6.4 and 4.1 units. What is the difference? | `6.4-4.1` | A difference of means. | 2.3 |
| D9 | A town of 24,000 adults is estimated at 38% support, with a margin of 3 points. How many adults does the lower end represent? | `0.35*24000` | Apply the lower percentage to the population. | 8400 |
| D10 | A sample proportion is 90 out of 150. What percentage is that? | `90/150*100` | The sample statistic, before any margin is applied. | 60 |

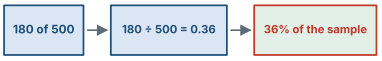
## Part 2 — Test-Style Questions (12 questions · hard and very hard)

Twelve questions, hardest last. Write your answer in the box — there are no options to choose from, so show the working that gets you there. Run the two-question test on every one: **was selection random?** (how far it generalises) and **was assignment random?** (whether cause may be claimed). They are independent.

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P1 | Hard | Researchers recorded the diets and blood pressure of 500 adults who had each chosen their own diet, and found that a high-salt diet was associated with higher blood pressure. State the best-supported conclusion.<br>  | **There is an association between the two, but the study cannot establish cause.** | Nobody was assigned a diet, so this is an **observational study** and only association follows.<br>Two common wrong answers assert a direction of cause the design cannot support. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P2 | Hard | In a study, 300 volunteers are randomly assigned to either a new tutoring method or the usual one, and the new method's group scores higher. State the best-supported conclusion.<br>  | **The new method causes higher scores for people like these volunteers.** | Random assignment supports a **causal claim**, but the volunteers were not randomly selected, so the finding cannot be stretched beyond people like them.<br>A common wrong answer is the classic overreach. |
| P3 | Hard | A study finds that towns with more public parks have lower obesity rates. Which is the most plausible confounding variable?<br>  | **The average household income of the town.** | A confounder must plausibly affect **both** quantities. Wealthier towns tend to have more parks and lower obesity.<br>Two common wrong answers name the two variables already in the study. |
| P4 | Hard | A researcher randomly selects 200 students from one school and surveys them about study habits. To which group may the results be generalised?<br>  | **All students at that school.** | The sample was drawn at random from that school, so the conclusion extends to **that school and no further**.<br>A common wrong answer is too cautious: random sampling is exactly what licenses the step from sample to school. |
| P5 | Hard | A survey of 400 people reports a margin of error of 5 percentage points. The researchers repeat it with 1,600 people, sampled the same biased way. Which improves?<br>  | **Precision only; the estimate is still centred on the wrong value.** | A larger sample narrows the margin of error, so the estimate becomes more **precise**.<br>But the same flawed selection means it is still **precisely wrong**. Size does not fix bias. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P6 | Very hard | Two studies are described. Study 1 randomly selects 500 adults from a city and surveys their exercise habits. Study 2 takes 500 volunteers and randomly assigns them to two exercise programmes. State the best-supported conclusion.<br>  <br>random SELECTION → you may generalise to the population<br>random ASSIGNMENT → you may claim cause and effect<br>neither one on its own gives you the other | **Study 1 supports generalisation to the city; Study 2 supports a causal claim for its participants.** | Random **selection** (Study 1) buys generalisation; random **assignment** (Study 2) buys cause.<br>A common wrong answer swaps them, which is the error the question is built around. |
| P7 | Very hard | A newspaper reports: 'Students who eat breakfast score higher on tests, so schools should provide breakfast.' The data came from a survey of students' existing habits. What is the main flaw?<br>the headline fails all three checks<br>how were they chosen? not at random × no<br>how were they assigned? not assigned at all × no<br>how far is it stretched? to every school × no | **The recommendation assumes cause from an observational study.** | The survey recorded what students **already did**, so it shows association only. Families who provide breakfast may differ in other ways.<br>The recommendation needs an experiment behind it. |
| P8 | Very hard | A clinical trial randomly assigns 300 patients, all recruited from one hospital, to a drug or a placebo. The drug group improves significantly more. State the best-supported conclusion.<br>a real experiment on a narrow group<br>random selection? no — one hospital only × no<br>random assignment? yes ✓ yes<br>so you may claim cause, for these patients ✓ yes | **The drug caused the improvement in this group of patients.** | Random assignment establishes **cause within the trial**.<br>Recruitment from a single hospital is not random selection from any wider population, so A over-reaches; C under-reads the assignment. |
| P9 | Very hard | A researcher wants to estimate the average commute time of all adults in a city and surveys every adult leaving one train station at 8 a.m. Which single change would most improve the study?<br>bias is about who was chosen, never how many<br>  <br>people who take the train commute differently from everyone else<br>a bigger biased sample is just a more precise wrong answer | **Select adults at random from the city's electoral register.** | The flaw is the **sampling frame**: train users are not all adults, and no sample size fixes that.<br>Only random selection from the whole population does. A common wrong answer makes a biased estimate more precise, not more accurate. |

| # | Level | Question | Answer | Solution |
|---|---|---|---|---|
| P10 | Very hard | In an experiment, 250 adults on diet A improved by a mean of 6.4 units and 250 on diet B by a mean of 4.1 units. Assignment was random. What is the difference in mean improvement?<br>  <br>random assignment is what lets you blame the diets | 2.3 | 6.4 − 4.1 = **2.3 units**.<br>Because assignment was random, this difference can reasonably be attributed to the diets. |
| P11 | Very hard | A town has 24,000 adults. A random sample estimates 38% support for a proposal, with a margin of error of 3 percentage points. Approximately how many adults does the lower end of the plausible interval represent?<br>  <br>take the lower end first, then scale it up | 8400 | Lower end = 38 − 3 = 35%.<br>0.35 × 24,000 = **8,400 adults**. |
| P12 | Very hard | Of 500 adults in an observational study, 180 reported a high-salt diet. What percentage of the sample is this?<br>  <br>a statistic about the sample, not yet about the population | 36 | 180/500 = 0.36 = **36%**.<br>Note this is a statistic about **the sample**; extending it to the population needs the sample to have been randomly selected. |

**The one habit to build.** Selection and assignment are **two different things** and answer two different questions. Most wrong answers on this topic are true statements about the wrong group, or causal claims from data where nobody assigned anything.

*All 18 worked examples and all 22 answers checked against the source and, where they are numeric, recomputed by verify13.py; every figure is drawn to the numbers in the question.*